Supplemental Material for "Prescribing finite-window reaction-time modes with a static fixed-budget Doi reactivity field"

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This Supplemental Material gives the bounded and natural-decay realizations of the two-particle Doi generator, proves compact-positive-time convergence of the required time and control mixed derivatives, and supplies the complete prescribed-modality proof. For each fixed finite dimension and fixed finite mode count, the construction uses a static, nonnegative lab-frame reactivity field of fixed centre-coordinate L^1 mass. A whole-window stationary signature, uniform over a compact family of positive slab weights, is first established for the free-exposure clock and then transferred to sufficiently small positive reaction budget. We also derive and evaluate a fixed-allocation sufficient threshold B_{cert}, distinct from both the uniform topology threshold $B_{\mathrm{top}}^{\mathrm{unif}}$ and the operational numerical crossing B_{op}, and record the off-lattice methods, sanity checks, configuration tables, classifier-sensitivity analysis, and supplementary figures.

S1. DOI MODEL AND ANALYTIC REALIZATIONS

For a fixed finite physical dimension $d \geq 2$, the longitudinal-slab symmetry reduction gives

$$\mathcal{Q}_d^{\mathrm{box}} = I_z \times I_\parallel \times \mathbb{T}_W^{d-1}, \qquad \mathcal{Q}_d^{\infty} = \mathbb{R} \times \mathbb{R} \times \mathbb{T}_W^{d-1}. \tag{S1}$$

With $x = (z, r_\parallel, r_\perp)$, equal particle diffusivities give the free forward operator

$$\mathcal{L}q = -\nabla\cdot(bq) + \nabla\cdot(\boldsymbol{D}\nabla q), \qquad \boldsymbol{D} = \mathrm{diag}(D/2, 2D, \ldots, 2D), \tag{S2}$$

where

$$b(x) = \big(-\gamma(z-\bar{z}), -\gamma r_\parallel, 0, \ldots, 0\big). \tag{S3}$$

For two independent particles with per-particle Fokker–Planck diffusivity D, the linear change of variables $Z = (X_1 + X_2)/2$, $R = X_1 - X_2$ gives diffusion entries $D/2$ and $2D$, respectively; identical longitudinal Ornstein–Uhlenbeck drifts give Eq. (S3). Comparison with the quotient-coordinate stochastic differential equations in Eqs. (S16)–(S18) therefore identifies

$$D = \varepsilon^2 D_0. \tag{S4}$$

Thus D_0 is the reference diffusion scale before the small-noise factor is applied; the physical diffusivity of either particle is D, not D_0. The bounded quotient has zero forward flux on its nonperiodic faces and periodic transverse coordinates. Natural decay on the unbounded quotient is specified by the reversible density

$$\pi(x) = Z_\pi^{-1}\exp\left[-\frac{\gamma(z-\bar{z})^2}{D} - \frac{\gamma r_\parallel^2}{4D}\right], \tag{S5}$$

including the normalized uniform torus factor.

Let $0 < a < W/2$, let $\chi_a(r)$ be the embedded minimum-image contact-ball indicator, and let normalized nonnegative longitudinal profiles obey $\phi_j \in L^\infty$ and $\int \phi_j(z)\,\mathrm{d}z = 1$. Put

$$V_j(z,r) = W^{-(d-1)}\chi_a(r)\phi_j(z), \qquad V_w = \sum_{j=1}^{J} w_j V_j, \qquad K_{B,w} = BV_w, \tag{S6}$$

where $w_j \geq 0$ and $\sum_j w_j = 1$; in Secs. S1–S3 the allocation is written w for the main text's $\boldsymbol{w}$. The installed centre-space catalyst amount is B for every w. The killed state and budget-rescaled reaction density are, with M_{V_w} the operator of multiplication by V_w,

$$q_{B,w}(t) = \mathrm{e}^{t(\mathcal{L}-BM_{V_w})}q_0, \qquad F_B(t,w) = \langle V_w, q_{B,w}(t)\rangle. \tag{S7}$$

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Proposition S1.1 (Analytic killed semigroups)**.** *On $\mathcal{Q}_d^{\mathrm{box}}$, the reflecting/periodic realization of $\mathcal{L}$ on $L^2(\mathcal{Q}_d^{\mathrm{box}})$ generates a positive analytic semigroup. On $\mathcal{Q}_d^{\infty}$, the natural-decay realization on $X_\pi = L^2(\pi^{-1}\mathrm{d}x)$ does the same. For real $B \geq 0$ and simplex control w, bounded multiplication by $-BV_w$ preserves the generator domain and yields a positive, mass-decreasing analytic semigroup. At positive time the state is entire in every finite affine complexification of the control amplitudes. On $\mathcal{Q}_d^{\infty}$, the observable pairing is continuous when $V_w \in L^2(\pi\mathrm{d}x) \cap L^\infty$.*

Proof. Set $q = \pi u$. The identity

$$\mathcal{L}(\pi u) = \pi \mathcal{G} u, \qquad \mathcal{G} = \pi^{-1}\nabla\cdot(\pi \boldsymbol{D}\nabla) \tag{S8}$$

follows from $b = \boldsymbol{D}\nabla \log \pi$. On $L^2(\pi\mathrm{d}x)$, the closed weighted form has

$$\langle u, \mathcal{G}v\rangle_{L^2(\pi)} = -\int (\nabla u)^{\mathsf{T}} \boldsymbol{D}\nabla v \,\pi\mathrm{d}x. \tag{S9}$$

It therefore realizes $\mathcal{G}$ as a nonpositive self-adjoint generator. Multiplication by π is a bounded isomorphism on the bounded quotient and a unitary map into X_π on the unbounded quotient. Bounded multiplication by $-BV_w$ preserves analyticity and the domain. The norm-convergent Dyson–Phillips series gives entire dependence on the finite control amplitudes. Positivity and mass decrease for real nonnegative data follow from the product formula, equivalently from Feynman–Kac [1, 2]. □

S2. UNIFORM COMPACT-POSITIVE-TIME MIXED DERIVATIVES

Choose an interior affine control chart $w(\theta) = \bar{w} + P\theta$ and a compact real control set Θ. Fix $\delta > 0$ such that each closed coordinate polydisc of radius δ about a point of Θ lies in the complex tube

$$\Theta_\delta^+ = \{\zeta \in \mathbb{C}^{J-1} : \mathrm{dist}_\infty(\zeta, \Theta) < 2\delta\}. \tag{S10}$$

Define

$$G(t,\theta) = \langle V_{w(\theta)}, \mathrm{e}^{t\mathcal{L}} q_0\rangle, \qquad v_{\infty,\delta} = \sup_{\zeta\in\Theta_\delta^+} \|V_{w(\zeta)}\|_\infty. \tag{S11}$$

Let $v_{*,\delta}$ be the supremum of $\|V_{w(\zeta)}\|_{L^2}$ on the bounded quotient and of $\|V_{w(\zeta)}\|_{L^2(\pi\mathrm{d}x)}$ on the unbounded quotient. Write κ_π for the bounded-quotient similarity bound and set $\kappa_\pi = 1$ on the unbounded quotient. Write X for $L^2(\mathcal{Q}_d^{\mathrm{box}})$ on the bounded quotient and for X_π on the unbounded one.

Theorem S2.1 (Uniform weak-reaction mixed-jet bridge)**.** *Fix $0 < \tau \leq T < \infty$, $0 \leq B \leq B_{\max}$, an integer $r \geq 0$, and a finite control multi-index α. The budget-rescaled density $F_B(t,\theta)$ in Eq.* (S7) *extends continuously to $F_0 = G$, and*

$$\begin{aligned}&\sup_{(t,\theta)\in[\tau,T]\times\Theta} |\partial_t^r \partial_\theta^\alpha (F_B - G)| \\ &\leq r!\,\alpha! \left(\frac{2}{\tau}\right)^r \delta^{-|\alpha|} v_{*,\delta}\kappa_\pi \left[\exp\left(\frac{3}{2}Bv_{\infty,\delta}T\right) - 1\right] \|q_0\|_X.\end{aligned} \tag{S12}$$

In particular, the left side is at most $BC_{r,\alpha}$ for a finite constant independent of $B \in [0, B_{\max}]$.

Proof. For complex time z with $\operatorname{Re} z > 0$, the norm-convergent Dyson expansion in the self-adjoint representation is

$$\mathrm{e}^{z(\mathcal{G}-BV_w)} = \mathrm{e}^{z\mathcal{G}} + \sum_{n=1}^{\infty} (-Bz)^n \int_{\Delta_n} \mathrm{e}^{z(1-s_1)\mathcal{G}} V_w \left[\prod_{k=1}^{n-1} \mathrm{e}^{z(s_k - s_{k+1})\mathcal{G}} V_w\right] \mathrm{e}^{zs_n\mathcal{G}}\,\mathrm{d}\boldsymbol{s}, \tag{S13}$$

where $\Delta_n = \{1 \geq s_1 \geq \cdots \geq s_n \geq 0\}$ has volume $1/n!$. Every free factor is contractive. Consequently,

$$\|\mathrm{e}^{z(\mathcal{L}-BV_w)} - \mathrm{e}^{z\mathcal{L}}\|_{X\to X} \leq \kappa_\pi\{\mathrm{e}^{B|z|\|V_w\|_\infty} - 1\}. \tag{S14}$$

The complex-bilinear observable remainder thus satisfies

$$\left|\left\langle V_{w(\zeta)}, [\mathrm{e}^{z(\mathcal{L}-BV_{w(\zeta)})} - \mathrm{e}^{z\mathcal{L}}]q_0\right\rangle\right| \leq v_{*,\delta}\kappa_\pi\{\mathrm{e}^{Bv_{\infty,\delta}|z|} - 1\}\|q_0\|_X. \tag{S15}$$

For $t \in [\tau, T]$, the disk $|z - t| \leq \tau/2$ remains in the open right half-plane and has $|z| \leq 3T/2$. Cauchy's estimate on this disk and on each radius-δ control polydisc proves Eq. (S12). Finally, $\mathrm{e}^{Bx} - 1 \leq Bx\mathrm{e}^{B_{\max}x}$ gives the linear bound. □

The estimate transfers any fixed finite collection of strict stationary-point neighbourhood, curvature, endpoint, and complement inequalities. It is asserted only on compact positive-time intervals and does not supply a useful numerical budget threshold.

S3. EXACT PRESCRIBED FINITE MODALITY: COMPLETE PROOF

This section proves the exact finite-mode statement used in the main article; Theorem S3.8 below is Theorem 1 of the main text. The construction is pointwise in a fixed finite dimension and a fixed finite mode count. Its two small parameters are ordered: first the narrow-noise parameter is fixed sufficiently small, and only then is the Doi budget taken sufficiently small. All stationary-point counts below are on the declared compact positive-time window.

S3.1. Exact quotient, stationary midpoint variance, and whole-window contact

Fix finite integers $d \geq 2$ and $m \geq 1$, a transverse period $W > 0$, a contact radius $0 < a < W/2$, and a compact positive-time interval $I = [\tau, T] \subset (0, \infty)$. On $\mathbb{R} \times \mathbb{R} \times \mathbb{T}_W^{d-1}$, let the longitudinal midpoint and relative coordinates solve

$$\mathrm{d}Z_t = -\gamma(Z_t - \bar{z})\,\mathrm{d}t + \varepsilon\sqrt{D_0}\,\mathrm{d}W_t, \tag{S16}$$

$$\mathrm{d}R_{\parallel,t} = -\gamma R_{\parallel,t}\,\mathrm{d}t + 2\varepsilon\sqrt{D_0}\,\mathrm{d}W_{\parallel,t}, \tag{S17}$$

$$\mathrm{d}R_{\perp,t} = 2\varepsilon\sqrt{D_0}\,\mathrm{d}W_{\perp,t} \quad (\mathrm{mod}\ W), \tag{S18}$$

where $D_0, \gamma > 0$. All Brownian drivers and all initial variables are mutually independent, and

$$Z_0 \sim N\left(z_0, \varepsilon^2 \frac{D_0}{2\gamma}\right), \qquad z_0 \neq \bar{z}, \tag{S19}$$

$$R_{\parallel,0} \sim N(r_{\parallel,0}, \varepsilon^2 u_0^2), \qquad 0 < u_0^2 < \frac{4D_0}{\gamma}, \tag{S20}$$

$$R_{\perp,0} \sim \mathrm{WN}(r_{\perp,0}, \varepsilon^2 \Sigma_{\perp,0}), \qquad \Sigma_{\perp,0} \succ 0. \tag{S21}$$

Here WN denotes the wrapped Gaussian on the transverse torus. The two strict variance inequalities $D_0/(2\gamma) < D_0/\gamma$ and $u_0^2 < 4D_0/\gamma$ are precisely the unbounded Gaussian conditions that place this initial law in the weighted state space used by Theorem S2.1; the wrapped factor is square integrable against the uniform torus measure.

The midpoint has mean and variance

$$\mu(t) = \bar{z} + (z_0 - \bar{z})\mathrm{e}^{-\gamma t}, \qquad \operatorname{Var} Z_t = \varepsilon^2 \frac{D_0}{2\gamma} \quad (t \geq 0). \tag{S22}$$

Thus only the variance, not the mean, is stationary. This constant variance is what produces a common-variance Gaussian mixture below.

Since μ' has constant sign, write $\operatorname{sgn}\mu' \in \{-1, 1\}$; fix a physical reference length $\ell_0 > 0$, and define

$$x(t) = \frac{\operatorname{sgn}(\mu')\,\mu(t)}{\ell_0}, \qquad v_0 := \inf_{t \in I} x'(t) > 0. \tag{S23}$$

For the present design $x'(t) = \gamma|z_0 - \bar{z}|\mathrm{e}^{-\gamma t}/\ell_0$, so $v_0 = \gamma|z_0 - \bar{z}|\mathrm{e}^{-\gamma T}/\ell_0$; the constant K of Lemma S3.6 is chosen of order $1/v_0$ below, which is why later windows require smaller ε. Choose target times and their dimensionless centres by

$$\tau < t_1 < \cdots < t_m < T, \qquad c_j = x(t_j). \tag{S24}$$

Then $c_1 < \cdots < c_m$, and the physical centre of slab j is $\operatorname{sgn}(\mu')\,\ell_0 c_j = \mu(t_j)$. In particular, reversing the trajectory orientation reverses the physical centres at the same time; no centre is held fixed under the coordinate reversal.

Write

$$r_*(t) = (r_{\parallel,0}\mathrm{e}^{-\gamma t}, r_{\perp,0}) \tag{S25}$$

and impose the whole-window, minimum-image contact margin

$$\sup_{t \in I} |r_*(t)|_{\mathrm{mi}} \leq a - \eta \quad \text{for some } \eta > 0. \tag{S26}$$

The word "whole-window" is essential: the condition is required on every point of I, not merely near the target times.

Lemma S3.1 (Whole-window differentiated contact tail)**.** *Let*

$$c_{d,\varepsilon}(t) = \mathbb{P}\{|R_t|_{\mathrm{mi}} < a\}. \tag{S27}$$

Under Eqs. (S17)–(S26)*, for $r = 0, 1, 2$ there are finite constants $C_{r,d}, N_{r,d}, q_d > 0$, independent of all sufficiently small ε, such that*

$$\|\partial_t^r (c_{d,\varepsilon} - 1)\|_{L^\infty(I)} \le C_{r,d}\varepsilon^{-N_{r,d}} \exp(-q_d/\varepsilon^2). \tag{S28}$$

Consequently, $c_{d,\varepsilon} \ge 1/2$ for sufficiently small ε, and the first two logarithmic derivatives of $c_{d,\varepsilon}$ are uniformly bounded on I and converge to zero faster than any power of ε.

Proof. In an unwrapped transverse chart the relative Gaussian has mean $r_*(t)$ and covariance $\varepsilon^2\Sigma_R(t)$, with longitudinal and transverse blocks

$$u_0^2 \mathrm{e}^{-2\gamma t} + \frac{2D_0}{\gamma}(1 - \mathrm{e}^{-2\gamma t}), \qquad \Sigma_{\perp,0} + 4D_0 t I_{d-1}. \tag{S29}$$

Because $I \subset (0, \infty)$, $u_0 > 0$, and $\Sigma_{\perp,0} \succ 0$, this covariance and its first two time derivatives are uniformly bounded on I, and its eigenvalues lie between two positive constants. The reverse triangle inequality and Eq. (S26) give

$$\inf_{t\in I} \inf_{|r|_{\mathrm{mi}}\ge a} d_{\mathrm{cyl}}(r, r_*(t)) \ge \eta. \tag{S30}$$

The assumption $a < W/2$ keeps the contact ball away from the transverse cut locus. Express the wrapped density as its Gaussian lattice-image sum. Every time derivative of order at most two is the same Gaussian multiplied by a polynomial in the scaled displacement and in ε^{-1}, with degree bounded independently of t. Equation (S30), the uniform covariance bounds, and a standard Gaussian tail estimate therefore give, after summing the lattice images,

$$\sup_{t\in I} \int_{|r|_{\mathrm{mi}}\ge a} |\partial_t^r p_{\varepsilon,t}(r)|\, \mathrm{d}r \le C_{r,d}\varepsilon^{-N_{r,d}} \mathrm{e}^{-q_d/\varepsilon^2}, \qquad r = 0, 1, 2. \tag{S31}$$

Differentiation under the integral is justified by the same integrable majorant and yields Eq. (S28). For small ε, its $r = 0$ case gives $c_{d,\varepsilon} \ge 1/2$, while

$$\begin{aligned} &|\partial_t \log c_{d,\varepsilon}| \le 2|\partial_t c_{d,\varepsilon}|, \\ &|\partial_t^2 \log c_{d,\varepsilon}| \le 2|\partial_t^2 c_{d,\varepsilon}| + 4|\partial_t c_{d,\varepsilon}|^2. \end{aligned} \tag{S32}$$

This proves the logarithmic-derivative statement. ☐

Fix $\rho > 0$, put

$$S_*^2 = \frac{D_0}{2\gamma} + \rho^2, \qquad \sigma = \frac{\varepsilon S_*}{\ell_0}, \tag{S33}$$

and use the normalized physical slabs

$$\phi_{j,\varepsilon}(z) = \frac{1}{\sqrt{2\pi}\,\varepsilon\rho} \exp\left[-\frac{(z - \mu(t_j))^2}{2\varepsilon^2\rho^2}\right]. \tag{S34}$$

Let $0 < w_* \le 1/m$ and

$$\mathcal{W}_{w_*} = \left\{ w = (w_1, \dots, w_m) : \sum_{j=1}^m w_j = 1, \quad w_j \ge w_* > 0 \right\}. \tag{S35}$$

For $w \in \mathcal{W}_{w_*}$, define the Doi killing field

$$K_{B,w,\varepsilon}(Z, R) = \mathbf{1}_{\{|R|_{\mathrm{mi}}<a\}} \frac{B}{W^{d-1}} \sum_{j=1}^m w_j \phi_{j,\varepsilon}(Z) = BV_{w,\varepsilon}(Z, R). \tag{S36}$$

Every slab has unit full longitudinal integral, so B is the same installed centre-space budget for every allocation.

Lemma S3.2 (Exact common-variance free-exposure factorization)**.** *Let $T_0(t)$ be the unkilled semigroup and q_0 the initial law above. The unit-budget free-exposure clock is exactly*

$$G_{\varepsilon,w}(t) := \langle V_{w,\varepsilon}, T_0(t)q_0\rangle = a_\varepsilon(t)H_{\sigma,w}(x(t)), \tag{S37}$$

$$a_\varepsilon(t) := \frac{c_{d,\varepsilon}(t)}{W^{d-1}\sqrt{2\pi}\,\varepsilon S_*}, \tag{S38}$$

$$H_{\sigma,w}(x) := \sum_{j=1}^{m} w_j \exp\left[-\frac{(x-c_j)^2}{2\sigma^2}\right]. \tag{S39}$$

The positive factor a_ε has uniformly bounded first two logarithmic time derivatives for all sufficiently small ε.

Proof. Independence factors the contact event from every longitudinal slab expectation. By Eq. (S22), convolution of $\phi_{j,\varepsilon}$ with the law of Z_t gives

$$\mathbb{E}[\phi_{j,\varepsilon}(Z_t)] = \frac{1}{\sqrt{2\pi}\,\varepsilon S_*}\exp\left[-\frac{(\mu(t)-\mu(t_j))^2}{2\varepsilon^2 S_*^2}\right]. \tag{S40}$$

Since $\mu(t)-\mu(t_j) = \operatorname{sgn}(\mu')\,\ell_0(x(t)-c_j)$, Eqs. (S33) and (S40) give Eqs. (S37)–(S39). The last assertion follows from Lemma S3.1, because the remaining factors in a_ε are independent of time. □

S3.2. The pure common-variance mixture

Set $J = x(I) = [\alpha,\beta]$. The targets are interior, so

$$\alpha < c_1 < \cdots < c_m < \beta. \tag{S41}$$

For $m \geq 2$, let

$$\Delta_j = c_{j+1} - c_j, \qquad \Delta_* := \min_{1\leq j<m} \Delta_j > 0. \tag{S42}$$

No empty minimum is used when $m = 1$.

Define

$$q_j(x) = w_j \mathrm{e}^{-(x-c_j)^2/(2\sigma^2)}, \qquad \pi_j(x) = \frac{q_j(x)}{H_{\sigma,w}(x)}, \qquad \bar{c}(x) = \sum_{j=1}^{m} \pi_j(x)c_j. \tag{S43}$$

The logarithmic slope $L_{\sigma,w} = \partial_x \log H_{\sigma,w}$ obeys the exact identities

$$L_{\sigma,w}(x) = \frac{\bar{c}(x)-x}{\sigma^2}, \tag{S44}$$

$$L'_{\sigma,w}(x) = \frac{\mathrm{Var}_{\pi(x)}(c)}{\sigma^4} - \frac{1}{\sigma^2}. \tag{S45}$$

Indeed, differentiation of q_j gives the first identity, while $\pi'_j = \pi_j(c_j - \bar{c})/\sigma^2$ gives the second.

The mode cap for univariate Gaussian mixtures is classical [3, 4]. We retain the direct multiplicity-counting proof: it makes the pure-mixture count of Theorem S3.5 exhaustive including multiple zeros, and it is the statement machine-checked in Sec. S6. The explicit root locations, curvature scales, and full-complement margins that the Doi transfer needs, and that the zero bound alone does not supply, come from Theorem S3.5 and Lemma S3.6 below.

Lemma S3.3 (Global extended-Chebyshev zero bound)**.** *For distinct real centres and positive weights, $H'_{\sigma,w}$ has at most $2m-1$ real zeros counted with multiplicity. Its real zero set is finite.*

Proof. Multiplication by a nowhere-zero function preserves both zeros and their multiplicities, and direct differentiation gives

$$\sigma^2 \mathrm{e}^{x^2/(2\sigma^2)} H'_{\sigma,w}(x) = \sum_{j=1}^{m} w_j(c_j - x)\mathrm{e}^{-c_j^2/(2\sigma^2)}\mathrm{e}^{c_j x/\sigma^2}. \tag{S46}$$

We prove the slightly more general claim that a nonzero function

$$P_m(x) = \sum_{j=1}^{m} (a_j + b_j x) e^{\lambda_j x}, \qquad \lambda_1 < \cdots < \lambda_m, \tag{S47}$$

with no identically zero summand has at most $2m - 1$ real zeros counted with multiplicity.

First, the zero set is finite. After multiplication by $e^{-\lambda_1 x}$, the first affine term dominates as $x \to -\infty$; the last affine-exponential term dominates as $x \to +\infty$. Each dominating affine factor has a fixed sign sufficiently far in its tail. Thus all zeros lie in a compact interval. A nonzero real-analytic function has only finitely many zeros there.

Now induct on m. The $m = 1$ case is the affine zero bound. For $m > 1$, put $Q = e^{-\lambda_1 x} P_m$. Two derivatives annihilate the first affine term. For every $j \geq 2$, the second derivative of $(a_j + b_j x) e^{(\lambda_j - \lambda_1) x}$ is a nonzero affine polynomial times the same exponential, and the remaining positive exponents are distinct. If a real-analytic function such as Q has $N \geq 2$ real zeros counted with multiplicity, generalized Rolle counting gives at least $N - 2$ zeros of its second derivative counted with multiplicity: differentiation reduces the multiplicity at each root, and ordinary Rolle supplies a root between each pair of distinct roots. Hence

$$N(Q) - 2 \leq N(Q'') \leq 2(m - 1) - 1, \tag{S48}$$

so $N(P_m) \leq 2m - 1$; the cases $N < 2$ are immediate. In Eq. (S46), every affine coefficient has nonzero slope $-w_j e^{-c_j^2/(2\sigma^2)}$, so the claim applies. □

Lemma S3.4 (Uniform adjacent isolation)**.** *For the fixed finite centre set and the compact weight family, there are* $C_{\mathrm{iso}}, q_{\mathrm{iso}} > 0$ *such that, for every* $j < m$*,*

$$\sup_{x \in [c_j, c_{j+1}]} \frac{\sum_{k \notin \{j, j+1\}} q_k(x)}{q_j(x) + q_{j+1}(x)} \leq C_{\mathrm{iso}} e^{-q_{\mathrm{iso}}/\sigma^2} \tag{S49}$$

for all sufficiently small σ*, uniformly in* w*. On* $[\alpha, c_1]$*, component* 1 *has the analogous one-component bound, and on* $[c_m, \beta]$*, component* m *does.*

Proof. For $x \in [c_j, c_{j+1}]$ and $k < j$,

$$(x - c_k)^2 - (x - c_j)^2 = (c_j - c_k)(2x - c_j - c_k) \geq (c_j - c_k)^2. \tag{S50}$$

For $k > j + 1$, comparison with c_{j+1} gives the corresponding positive lower bound. Since $w_k/w_j \leq 1/w_*$, summation over the finite centre set proves Eq. (S49). The same difference-of-squares comparison proves the two tail statements. The posterior mean and variance of the full mixture therefore differ from those of the isolated adjacent pair by $O(e^{-q/\sigma^2})$, uniformly in the gap, the allocation, and sufficiently small σ. □

Theorem S3.5 (Exact topology of the pure mixture)**.** *Uniformly for* $w \in \mathcal{W}_{w_*}$*, every sufficiently small* $\sigma > 0$ *gives exactly* m *simple local maxima and* $m - 1$ *simple local minima of* $H_{\sigma,w}$ *on* J*, ordered alternately, with nonzero endpoint derivatives. The maximum near* c_j*, denoted* $p_j(\sigma, w)$*, satisfies*

$$p_j = c_j + O(e^{-q/\sigma^2}), \qquad L'_{\sigma,w}(p_j) = -\sigma^{-2} + o(\sigma^{-2}). \tag{S51}$$

For $m \geq 2$*, define the weighted equality point*

$$s_j(\sigma, w) = \frac{c_j + c_{j+1}}{2} + \frac{\sigma^2}{\Delta_j} \log \frac{w_j}{w_{j+1}}. \tag{S52}$$

The minimum in the j*-th gap, denoted* $r_j(\sigma, w)$*, satisfies*

$$r_j = s_j + O(\sigma^4), \tag{S53}$$

$$L'_{\sigma,w}(r_j) = \frac{\Delta_j^2}{4\sigma^4} + O(\sigma^{-2}). \tag{S54}$$

All remainders are uniform on the compact weight set.

Proof. Fix a constant $A>1$. On $|x-c_j|\le A\sigma^2$, one-component isolation gives

$$L_{\sigma,w}(x)=\frac{c_j-x}{\sigma^2}+O(\sigma^{-2}\mathrm{e}^{-q/\sigma^2}), \tag{S55}$$

$$L'_{\sigma,w}(x)=-\sigma^{-2}+O(\sigma^{-4}\mathrm{e}^{-q/\sigma^2}). \tag{S56}$$

At $c_j-\sigma^2$ the slope is positive, at $c_j+\sigma^2$ it is negative, and it is strictly decreasing between them. Hence there is one simple maximum with the location and derivative in Eq. (S51).

For a gap, the isolated adjacent odds are exactly

$$\frac{q_{j+1}(x)}{q_j(x)}=\exp\left[\frac{\Delta_j(x-s_j)}{\sigma^2}\right]. \tag{S57}$$

Put $h_j=(\log 9)/\Delta_j$ and

$$C_j=[s_j-h_j\sigma^2,s_j+h_j\sigma^2]. \tag{S58}$$

At the left and right endpoints the adjacent posterior masses are, respectively, $(9/10,1/10)$ and $(1/10,9/10)$. By Lemma S3.4, after reducing a common σ_0, both adjacent posterior masses are bounded below throughout C_j, and

$$\mathrm{Var}_{\pi(x)}(c)\ge\kappa_0\Delta_*^2,\qquad L'_{\sigma,w}(x)\ge\frac{\kappa_v}{\sigma^4}>0\quad(x\in C_j) \tag{S59}$$

for constants independent of j,w,σ. At the two endpoints of C_j, Eq. (S44) gives, for sufficiently small σ,

$$L_{\sigma,w}(s_j-h_j\sigma^2)\le-\frac{\Delta_j}{5\sigma^2},\qquad L_{\sigma,w}(s_j+h_j\sigma^2)\ge\frac{\Delta_j}{5\sigma^2}. \tag{S60}$$

Thus C_j contains exactly one simple minimum root.

At s_j, the adjacent posterior masses are equal and the adjacent posterior mean is $(c_j+c_{j+1})/2$. Hence

$$L_{\sigma,w}(s_j)=-\frac{1}{\Delta_j}\log\frac{w_j}{w_{j+1}}+O(\sigma^{-2}\mathrm{e}^{-q/\sigma^2})=O(1) \tag{S61}$$

uniformly in w. The mean-value theorem and Eq. (S59) give $r_j-s_j=O(\sigma^4)$. At r_j, the two adjacent posterior masses are $1/2+O(\sigma^2)$; substitution in Eq. (S45) proves Eq. (S54).

For small σ, the constructed peak and gap neighbourhoods are pairwise disjoint and contained in (α,β). They contain $m+(m-1)=2m-1$ distinct simple stationary points. The global bound in Lemma S3.3 excludes every additional real stationary point, including multiple ones. Tail isolation and the fixed separations $c_1-\alpha>0$ and $\beta-c_m>0$ give

$$L_{\sigma,w}(\alpha)>0,\qquad L_{\sigma,w}(\beta)<0, \tag{S62}$$

so neither endpoint is stationary. When $m=1$, there are no gaps or valley definitions: the exact identity $L_{\sigma,w}(x)=(c_1-x)/\sigma^2$ gives one maximum and both endpoint signs directly. □

S3.3. Full posterior-sector certificate and a slow positive factor

The zero bound above does not apply after multiplication by a nonconstant positive factor. We therefore certify the complete complement of small peak and valley boxes.

Lemma S3.6 (Posterior-sector certificate)**.** *Fix $K>0$. There are constants $A_\mathrm{p},A_\mathrm{v},C_\mathrm{p},C_\mathrm{v},\kappa_\mathrm{p},\kappa_\mathrm{v}>0$ and $\sigma_K>0$, depending only on the fixed centre set, J, w_*, and K, such that, for every $0<\sigma<\sigma_K$ and every admissible w, the boxes*

$$P_j=[c_j-A_\mathrm{p}\sigma^2,c_j+A_\mathrm{p}\sigma^2], \tag{S63}$$

$$V_j=[r_j-A_\mathrm{v}\sigma^4,r_j+A_\mathrm{v}\sigma^4]\quad(j<m) \tag{S64}$$

are pairwise disjoint and contained in (α,β), and the following hold:

$$|L_{\sigma,w}|\le C_\mathrm{p},\qquad L'_{\sigma,w}\le-\kappa_\mathrm{p}\sigma^{-2}\qquad \textit{on } P_j, \tag{S65}$$

$$|L_{\sigma,w}|\le C_\mathrm{v},\qquad L'_{\sigma,w}\ge\kappa_\mathrm{v}\sigma^{-4}\qquad \textit{on } V_j. \tag{S66}$$

At the left and right box boundaries, respectively, the slope signs are

$$\begin{array}{c|cc} & \textit{left boundary} & \textit{right boundary} \\ \hline P_j & L_{\sigma,w} \geq 2K & L_{\sigma,w} \leq -2K \\ V_j & L_{\sigma,w} \leq -2K & L_{\sigma,w} \geq 2K. \end{array} \tag{S67}$$

On the entire complement in J of the open boxes,

$$|L_{\sigma,w}(x)| \geq 2K, \tag{S68}$$

with the exhaustive alternating sign order

$$+\ P_1\ -\ V_1\ +\ P_2\ -\ \cdots\ +\ P_m\ -\,. \tag{S69}$$

For $m = 1$, every valley clause is omitted.

Proof. Choose $A_\mathrm{p} \geq 4K$. The uniform expansions (S55)–(S56) give, after reducing σ_K, Eq. (S65) and the peak row of Eq. (S67). The convex-hull property $\bar{c}(x) \in [c_1, c_m]$ gives the full outer-tail estimates without any local expansion:

$$\begin{aligned} x \leq c_1 - A_\mathrm{p}\sigma^2 &\Longrightarrow L_{\sigma,w}(x) \geq A_\mathrm{p}, \\ x \geq c_m + A_\mathrm{p}\sigma^2 &\Longrightarrow L_{\sigma,w}(x) \leq -A_\mathrm{p}. \end{aligned} \tag{S70}$$

Fix a gap and abbreviate $\Delta = \Delta_j$, $s = s_j$. The right adjacent posterior mass is

$$p(x) = \frac{q_{j+1}(x)}{q_j(x) + q_{j+1}(x)} = \frac{1}{1 + \exp[-\Delta(x-s)/\sigma^2]}. \tag{S71}$$

On the crossover interval C_j in Eq. (S58), its edge odds are 1/9 and 9, not exponentially small or large. Both adjacent masses are therefore bounded below throughout C_j, and Eq. (S59) holds there. Conversely, the fixed finite centre diameter bounds the posterior variance, so $|L'| \leq C\sigma^{-4}$ on C_j.

The pure root satisfies $L(r_j) = 0$. Choose A_v large enough that $\kappa_v A_\mathrm{v} \geq 2K$. Since $A_\mathrm{v}\sigma^4 = o(\sigma^2)$, the valley box lies in C_j for small σ. Integrating the lower and upper bounds for L' from r_j proves Eq. (S66), the valley row of Eq. (S67), and the required negative and positive bounds on $C_j \setminus \operatorname{int} V_j$.

It remains to cover both outer sectors of every gap; this is the step that prevents an unproved dominance claim at a crossover edge. Consider first

$$[c_j + A_\mathrm{p}\sigma^2, s_j - h_j\sigma^2] \tag{S72}$$

and split it at $c_j + \Delta_j/4$, which lies in the stated interval for all sufficiently small σ, uniformly in w. On the portion nearer c_j, every other component is exponentially small relative to component j, so

$$\bar{c}(x) - x \leq -\tfrac{1}{2}A_\mathrm{p}\sigma^2, \qquad L_{\sigma,w}(x) \leq -A_\mathrm{p}/2 \leq -2K. \tag{S73}$$

On the remaining portion, monotonicity of Eq. (S71) and the left crossover edge give $p(x) \leq 1/10$, while $x - c_j \geq \Delta_j/4$. The adjacent-pair mean then obeys

$$c_j + \Delta_j p(x) - x \leq \frac{\Delta_j}{10} - \frac{\Delta_j}{4} = -\frac{3\Delta_j}{20}. \tag{S74}$$

The nonadjacent correction is exponentially small by Lemma S3.4; hence this sector also has $L \leq -2K$ for small σ, including its 1/9-odds edge.

Symmetrically, split $[s_j + h_j\sigma^2, c_{j+1} - A_\mathrm{p}\sigma^2]$ at $c_{j+1} - \Delta_j/4$. On the portion nearer the crossover, $p(x) \geq 9/10$ and $c_{j+1} - x \geq \Delta_j/4$, giving the positive analogue of Eq. (S74); on the portion nearer c_{j+1}, one-component isolation gives $L \geq A_\mathrm{p}/2 \geq 2K$. Thus the right outer sector has $L \geq 2K$.

The fixed centre and endpoint separations make all boxes disjoint and interior for one common sufficiently small σ_K. The outer tails, all peak boundaries, both outer sectors of every gap, and every crossover-minus-valley sector cover every point of the box complement. This proves Eqs. (S68)–(S69). When $m = 1$, the same result follows directly from $L = (c_1 - x)/\sigma^2$ and Eq. (S70). □

Theorem S3.7 (Slow positive factors preserve the exact topology)**.** *Let $x \in C^2(I)$ be strictly increasing with $\inf_I x' = v_0 > 0$. Let $a_\sigma \in C^2(I)$ be positive and assume*

$$M_0 := \sup_{\sigma<\sigma_0} \|\partial_t \log a_\sigma\|_{L^\infty(I)} < \infty, \qquad M_1 := \sup_{\sigma<\sigma_0} \|\partial_t^2 \log a_\sigma\|_{L^\infty(I)} < \infty. \tag{S75}$$

Then, uniformly for $w \in \mathcal{W}_{w_*}$, all sufficiently small $\sigma > 0$ give exactly m nondegenerate maxima and $m-1$ nondegenerate minima of

$$F_{\sigma,w}(t) = a_\sigma(t) H_{\sigma,w}(x(t)) \tag{S76}$$

on I, ordered alternately, with no stationary endpoint. Relative to the pure roots in the x coordinate,

$$x(t^{\max}_{j,\sigma}) - p_j(\sigma, w) = O(\sigma^2), \tag{S77}$$
$$x(t^{\min}_{j,\sigma}) - r_j(\sigma, w) = O(\sigma^4), \tag{S78}$$

uniformly in w. Thus

$$x(t^{\min}_{j,\sigma}) = s_j(\sigma, w) + O(\sigma^4). \tag{S79}$$

Proof. Put

$$b_\sigma = \partial_t \log a_\sigma, \qquad D_{\sigma,w}(t) = \partial_t \log F_{\sigma,w}(t) = b_\sigma(t) + x'(t) L_{\sigma,w}(x(t)). \tag{S80}$$

Since $F_{\sigma,w} > 0$, its stationary points are precisely the zeros of $D_{\sigma,w}$, and

$$D'_{\sigma,w} = b'_\sigma + x'' L_{\sigma,w} + (x')^2 L'_{\sigma,w}. \tag{S81}$$

Choose $K > (M_0 + 1)/v_0$ and apply Lemma S3.6. On the full complement of the open boxes, the term $x'L$ strictly dominates b_σ, so D has the alternating signs in Eq. (S69) and has no complement zero.

On a peak box, Eqs. (S65) and (S81) give

$$D'_{\sigma,w} \le M_1 + \|x''\|_\infty C_{\mathrm{p}} - v_0^2 \kappa_{\mathrm{p}} \sigma^{-2} < 0 \tag{S82}$$

for small σ. Its boundary signs give exactly one zero. At that zero $F'' = FD' < 0$, hence the root is a nondegenerate maximum. Similarly, on a valley box,

$$D'_{\sigma,w} \ge -M_1 - \|x''\|_\infty C_{\mathrm{v}} + v_0^2 \kappa_{\mathrm{v}} \sigma^{-4} > 0, \tag{S83}$$

so its unique zero is a nondegenerate minimum. The certified tail signs exclude endpoint stationary points.

At a slow-factor root, Eq. (S80) gives $|L| \le M_0/v_0$. The pure peak root has $L(p_j) = 0$, and the uniform negative $L' = \Theta(\sigma^{-2})$ on its box implies the peak displacement in Eq. (S77). The uniform positive $L' = \Theta(\sigma^{-4})$ on a valley box gives Eq. (S78). Combining the latter with Eq. (S53) proves Eq. (S79). Every constant is uniform over the compact allocation family. □

S3.4. Fixed-ε Doi transfer and theorem boundary

For $B > 0$, let

$$f_{B,\varepsilon,w}(t) = \mathbb{E}_0\left[K_{B,w,\varepsilon}(Z_t, R_t) \exp\left(-\int_0^t K_{B,w,\varepsilon}(Z_s, R_s)\,\mathrm{d}s\right)\right] \tag{S84}$$

be the Doi reaction-time density and set

$$\mathcal{F}_{B,\varepsilon,w} = f_{B,\varepsilon,w}/B. \tag{S85}$$

This is budget-rescaling, not normalization to unit time integral.

Theorem S3.8 (Exact m Doi modes for fixed finite (d,m))**.** *Fix all data in Eqs.* (S16)–(S36)*, including the stationary midpoint variance, mutual independence, the strict weighted-space variance inequalities, target times strictly inside I, consistently oriented physical centres, normalized slabs of physical width $\varepsilon\rho$, the whole-window contact margin, and the compact allocation family. Then there is $\varepsilon_0 > 0$, depending on these fixed data, such that for every $0 < \varepsilon < \varepsilon_0$ and every $w \in \mathcal{W}_{w_*}$, the continuum free-exposure clock $G_{\varepsilon,w}$ has exactly m nondegenerate maxima and $m-1$ nondegenerate minima on I, ordered alternately, and no stationary endpoint.*

For each fixed such ε and allocation set $\mathcal{A} \subseteq \mathcal{W}_{w_*}$, define

$$B_{\mathrm{top}}(\varepsilon;\mathcal{A}) := \sup\big\{b > 0 : \mathcal{F}_{\beta,\varepsilon,w} \text{ has that complete signature on } I \\ \text{for every } w \in \mathcal{A} \text{ and every } 0 < \beta < b\big\}. \tag{S86}$$

Then $B^{\mathrm{unif}}_{\mathrm{top}}(\varepsilon) := B_{\mathrm{top}}(\varepsilon;\mathcal{W}_{w_*}) > 0$. Consequently $\mathcal{F}_{B,\varepsilon,w}$ has the stated signature for every $0 < B < B^{\mathrm{unif}}_{\mathrm{top}}(\varepsilon)$ and every admissible allocation. Multiplication by $B > 0$ changes no stationary point, so the statement also holds for the physical density $f_{B,\varepsilon,w}$.

Proof. In Lemma S3.2, take $a_\sigma = a_\varepsilon$, with $\sigma = \varepsilon S_*/\ell_0$. Lemma S3.1 supplies Eq. (S75); hence Theorem S3.7 proves the exact stationary signature of $G_{\varepsilon,w}$, uniformly over the compact allocation family.

Fix one $0 < \varepsilon < \varepsilon_0$. The ordered simple roots depend continuously on w by the implicit-function theorem. Their finitely many graphs over the compact set $\mathcal{W}_{w_*}$ are compact and disjoint. Continuity of the second time derivative gives pairwise disjoint root tubes on which G'' has a uniform nonzero sign and magnitude. After shrinking the tubes if necessary, G' has fixed opposite signs at the two boundaries of each tube. On the compact complement in $I \times \mathcal{W}_{w_*}$, $|G'|$ has a positive minimum; the two endpoint slopes have positive uniform absolute minima as well.

For this fixed ε, $V_{w,\varepsilon}$ is bounded uniformly in w, and Eqs. (S19)–(S21) put q_0 in the required weighted state space. Because $w_j \geq w_* > 0$, the compact allocation set lies strictly inside the simplex and admits the affine control chart and a complex tube required by Theorem S2.1. (If the set is the singleton $w_j = 1/m$, the same conclusion is the corresponding pointwise case.) Applying that theorem with time orders $0, 1, 2$ and no control derivative gives

$$\sup_{w \in \mathcal{W}_{w_*}} \|\mathcal{F}_{B,\varepsilon,w} - G_{\varepsilon,w}\|_{C^2(I)} \longrightarrow 0 \qquad (B \downarrow 0). \tag{S87}$$

Choose $b_\varepsilon > 0$ so that the C^1 error is smaller than all boundary, complement, and endpoint margins and the second-derivative error is smaller than the root-tube curvature margin. In each tube, $\mathcal{F}'$ has opposite boundary signs and $\mathcal{F}''$ keeps the sign of G'', so it has exactly one root of the same type. The complement and endpoints contain none. Thus b_ε belongs to the defining set in Eq. (S86), so $B^{\mathrm{unif}}_{\mathrm{top}}(\varepsilon) \geq b_\varepsilon > 0$. This proves the complete finite-window signature. ☐

Remark S3.9 (Quantifiers, saturation, and scope)**.** *The quantifier order is*

$$\text{fix } (d, m, \text{all geometric and probabilistic data}) \implies 0 < \varepsilon < \varepsilon_0 \implies 0 < B < B^{\mathrm{unif}}_{\mathrm{top}}(\varepsilon). \tag{S88}$$

There is no asserted lower bound for $B^{\mathrm{unif}}_{\mathrm{top}}(\varepsilon)$ uniform as $\varepsilon \downarrow 0$, and no interchange of the two limits. The construction is not uniform in d or m, does not provide one geometry with arbitrarily many modes, does not count stationary points outside I, and gives no process-level event-mass floor. It treats normalized longitudinal Gaussian slabs, not arbitrary localized catalyst patches.

Moreover, Lemma S3.1 shows that the relative contact factor is asymptotically saturated on the theorem window:

$$c_{d,\varepsilon} = 1 + O(\varepsilon^{-N} \mathrm{e}^{-q/\varepsilon^2}) \quad \text{in } C^2(I). \tag{S89}$$

Thus the exact Doi embedding is rigorous, but the modal basis is created by the ordered narrow slabs sampled by the monotone constant-variance midpoint; the relative encounter factor is a slow perturbation in this asymptotic family. Nontrivial contact, a useful common positive budget, deterministic finite-parameter complement certificates, survival, and event-basin masses remain separate continuum-validation requirements rather than consequences of Theorem S3.8.

S3.5. A fixed-allocation sufficient budget

Let $B_{\mathrm{top}}(\varepsilon;\mathcal{A})$ denote the connected topology-retention threshold for an allocation set $\mathcal{A}$: the supremum of all $b > 0$ for which the complete stationary signature holds for every allocation in $\mathcal{A}$ and every budget in $(0, b)$. Theorem S3.8 proves $B_{\mathrm{top}}(\varepsilon;\mathcal{W}_{w_*}) > 0$ but obtains it by compactness. This subsection instead fixes one interior allocation w and constructs a computable sufficient budget $B_{\mathrm{cert}}(\varepsilon; w, \{U_i\})$. It satisfies

$$B_{\mathrm{cert}}(\varepsilon; w, \{U_i\}) \leq B_{\mathrm{top}}(\varepsilon; \{w\}), \tag{S90}$$

but is not a bound for the threshold uniform over a larger allocation family unless all constants are made uniform over that family. It must also not be identified with the operational finite-sample classifier crossing $B_{\rm op}$ reported in the numerical campaign. Two quantitative routes are available. The route through Theorem S2.1 — Cauchy–Schwarz in the weighted space $X_\pi = L^2(\pi^{-1}\mathrm{d}x)$ — is completely general but pays the weighted norm $\|q_0\|_X \sim \mathrm{e}^{\gamma(z_0-\bar{z})^2/(\varepsilon^2 D_0)}$, which destroys the numerical value of the resulting threshold; it is retained for comparison as Remark S3.15. The main result, Proposition S3.14, avoids the weighted norm altogether: it bounds the observable remainder by a Dyson expansion in the flat (forward, probabilistic) pairing, estimated leg by leg on the explicit complex-time Mehler kernels of the slab design, and recovers the two time derivatives required by the signature argument through a Cauchy estimate on disks of ε-scaled radius r_0. Every constant on the flat Dyson-transfer side of the sharpened threshold is explicit and grows at most polynomially in ε^{-1}; the signature margins themselves may be exponentially small. No weighted-norm exponential is introduced: the price is the radius $r_0 \propto \varepsilon$ in place of $\tau/2$ — a polynomial-in-ε Cauchy cost rather than a weighted-norm exponential. The sharpened route uses the product Ornstein–Uhlenbeck/Ornstein–Uhlenbeck/torus structure of the free process, the Gaussian slabs with the compactly supported contact indicator, and the Gaussian initial law [Eqs. (S16)–(S36)]; it is a proposition about this slab design, not a replacement for the abstract Theorem S2.1. Remark S3.16 evaluates both routes at a worked anchor and at two secondary parameter sets, using the same relative-coordinate initial law as the off-lattice campaign. The comparison with $B_{\rm op}$ is diagnostic only.

Setting and notation. Throughout this subsection the data of Theorem S3.8 are fixed on the unbounded natural-decay quotient $\mathcal{Q}_d^\infty$ of Eq. (S1), with the allocation family the singleton $\mathcal{W} = \{w\}$ for one fixed interior allocation w (the pointwise case recorded in the proof of Theorem S3.8). We write $\langle\cdot,\cdot\rangle$ for the flat (Lebesgue) pairing of a bounded function against an integrable density on $\mathcal{Q}_d^\infty$, so that the free-exposure clock and the budget-rescaled Doi density of Eqs. (S37) and (S85) are $G(t) = \langle V_{w,\varepsilon}, \mathrm{e}^{t\mathcal{L}} q_0\rangle$ and $\mathcal{F}(t) = \mathcal{F}_{B,\varepsilon,w}(t) = \langle V_{w,\varepsilon}, \mathrm{e}^{t(\mathcal{L}-BV_{w,\varepsilon})} q_0\rangle$ [Eq. (S7)]. Put $v_\infty = \|V_{w,\varepsilon}\|_\infty$ and define the observable remainder at complex time,

$$\mathcal{R}_B(z) = \big\langle V_{w,\varepsilon}, \big(\mathrm{e}^{z\mathcal{L}} - \mathrm{e}^{z(\mathcal{L}-BV_{w,\varepsilon})}\big) q_0\big\rangle, \qquad \mathrm{Re}\, z > 0, \tag{S91}$$

so that $\mathcal{R}_B(t) = G(t) - \mathcal{F}(t)$ for real $t > 0$.

Why the weighted norm appears, and how it is avoided. The route of Theorem S2.1 estimates $\mathcal{R}_B$ by an operator bound plus Cauchy–Schwarz in X_π [Eqs. (S14)–(S15)]; since q_0 is centred at $z_0 \neq \bar{z}$ while π is centred at $\bar{z}$ with variance $O(\varepsilon^2)$, the factor $\|q_0\|_X$ is exponentially large in $1/\varepsilon^2$. The weighted pairing is forced there because the complex-time free semigroup is a bounded analytic family only in the self-adjoint representation: on flat Lebesgue spaces the complex-time Ornstein–Uhlenbeck semigroup is not a bounded operator family (the complexified drift displaces mass by an imaginary amount proportional to the starting point, and the flat operator norm diverges). The repair rests on two observations. First, at real time no weighted norm is needed at all: by Feynman–Kac [Eq. (S84)] and $0 \le \int_0^t V(X_s)\,\mathrm{d}s \le v_\infty t$ pathwise, $0 \le G(t) - \mathcal{F}(t) \le (\mathrm{e}^{Bv_\infty t} - 1)G(t)$; the whole difficulty is the two *time derivatives* required by the signature argument. Second, derivatives can be recovered by Cauchy's estimate on disks $|z - t| \le r_0$ with $r_0 \propto \varepsilon$, because every leg of a flat Dyson chain starts either from the initial law or from a potential node, and both confine the leg's starting point to an $O(\varepsilon)$ neighbourhood of an $O(1)$ centre; with $|\,\mathrm{Im}\, z| \le r_0 = O(\varepsilon)$, the imaginary-displacement penalties are linear in the leg durations and total $O(1)$ per chain, uniformly in the chain order (Lemma S3.11). No operator bound on a flat complex semigroup is ever used: the weighted-space theory of Proposition S1.1 enters only *qualitatively*, to guarantee that $z \mapsto \mathcal{R}_B(z)$ is analytic and that its Dyson series converges, while every quantitative estimate is performed on explicit finite-dimensional complex Gaussian integrals (Proposition S3.13).

Kernels. For $\upsilon \in \mathbb{C}$ with $\mathrm{Re}\,\upsilon > 0$ write $\mathcal{N}(u;\upsilon) = \mathrm{e}^{-u^2/(2\upsilon)}/\sqrt{2\pi\upsilon}$ (principal square root). Over a real duration $u > 0$ the free transition density factorizes over the coordinate blocks of Eqs. (S16)–(S18): started at $y = (y_Z, y_\parallel, y_\perp)$,

$$\begin{aligned} p_u(x\,|\,y) = {}& \mathcal{N}\big(x_Z - \bar{z} - (y_Z - \bar{z})\mathrm{e}^{-\gamma u}; \varepsilon^2 v_Z(u)\big)\, \mathcal{N}\big(x_\parallel - y_\parallel \mathrm{e}^{-\gamma u}; \varepsilon^2 v_\parallel(u)\big) \\ & \times \sum_{k\in\mathbb{Z}^{d-1}} \prod_{i=1}^{d-1} \mathcal{N}\big(x_{\perp,i} - y_{\perp,i} + k_i W; \varepsilon^2 v_\perp(u)\big), \end{aligned} \tag{S92}$$

where the lattice sum wraps the transverse block onto the torus and

$$v_Z(\zeta) = \frac{D_0}{2\gamma}\big(1 - \mathrm{e}^{-2\gamma\zeta}\big), \qquad v_\parallel(\zeta) = \frac{2D_0}{\gamma}\big(1 - \mathrm{e}^{-2\gamma\zeta}\big), \qquad v_\perp(\zeta) = 4D_0\zeta, \tag{S93}$$

consistent with Eqs. (S22) and (S29). The mean multipliers $\mathrm{e}^{-\gamma\zeta}$ (blocks Z, $\parallel$) and 1 (block $\perp$) and the variances (S93) are entire in the duration; their values at $\zeta = zh$, $\mathrm{Re}\, z > 0$, $h \in (0, 1]$, define the *complex-leg kernels* used below (well defined by Lemma S3.11(a)).

Lemma S3.10 (Complex Gaussian domination)**.** *Let $\upsilon \in \mathbb{C}$ with $\operatorname{Re}\upsilon > 0$, write $\vartheta = \arg\upsilon \in (-\pi/2, \pi/2)$ and $1/(2\upsilon) = \mathfrak{a} - \mathrm{i}\mathfrak{b}$, so that $\mathfrak{a} = \cos\vartheta/(2|\upsilon|) > 0$ and $\mathfrak{b} = \sin\vartheta/(2|\upsilon|)$. Let $\mathfrak{m} = \mathfrak{m}_R + \mathrm{i}\mathfrak{m}_I \in \mathbb{C}$. Then for every $\lambda \in (0,1)$ and all $x \in \mathbb{R}$,*

$$\big|\mathcal{N}(x - \mathfrak{m}; \upsilon)\big| \le \kappa(\lambda, \vartheta)\, g_{s^2}(x - \mathfrak{m}_R)\, \exp\!\big(P_\lambda\, \mathfrak{m}_I^2\big), \tag{S94}$$

where g_{s^2} is the unit-mass real Gaussian density of variance $s^2 = 1/(2\mathfrak{a}(1-\lambda))$ and

$$\kappa(\lambda, \vartheta) = \big[(1-\lambda)\cos\vartheta\big]^{-1/2}, \qquad P_\lambda = \mathfrak{a} + \frac{\mathfrak{b}^2}{\lambda\mathfrak{a}} = \Big(1 + \frac{\tan^2\vartheta}{\lambda}\Big)\frac{\cos^2\vartheta}{2\operatorname{Re}\upsilon} \le \Big(1 + \frac{\tan^2\vartheta}{\lambda}\Big)\frac{1}{2\operatorname{Re}\upsilon}. \tag{S95}$$

If $\mathfrak{m}_I = 0$, the bound holds with $\lambda = 0$, $\kappa = (\cos\vartheta)^{-1/2}$, and no penalty factor.

Proof. $|\mathcal{N}(x - \mathfrak{m}; \upsilon)| = |2\pi\upsilon|^{-1/2}\exp\{-\operatorname{Re}[(x-\mathfrak{m})^2/(2\upsilon)]\}$. With $\xi = x - \mathfrak{m}_R$, $(x-\mathfrak{m})^2 = \xi^2 - \mathfrak{m}_I^2 - 2\mathrm{i}\xi\mathfrak{m}_I$, hence $\operatorname{Re}[(x-\mathfrak{m})^2(\mathfrak{a} - \mathrm{i}\mathfrak{b})] = \mathfrak{a}\xi^2 - \mathfrak{a}\mathfrak{m}_I^2 - 2\mathfrak{b}\mathfrak{m}_I\xi$ and

$$|\mathcal{N}| = |2\pi\upsilon|^{-1/2}\exp\!\big(-\mathfrak{a}\xi^2 + 2\mathfrak{b}\mathfrak{m}_I\xi + \mathfrak{a}\mathfrak{m}_I^2\big).$$

The cross term carries no useful sign — $\mathfrak{b}$ and $\mathfrak{m}_I$ may each take either sign — so it is bounded in absolute value by Young's inequality, $|2\mathfrak{b}\mathfrak{m}_I\xi| \le \lambda\mathfrak{a}\xi^2 + \mathfrak{b}^2\mathfrak{m}_I^2/(\lambda\mathfrak{a})$, giving $|\mathcal{N}| \le |2\pi\upsilon|^{-1/2}\mathrm{e}^{-(1-\lambda)\mathfrak{a}\xi^2}\,\mathrm{e}^{P_\lambda\mathfrak{m}_I^2}$. The Gaussian factor equals $\sqrt{2\pi s^2}\, g_{s^2}(\xi)$ with $s^2 = 1/(2\mathfrak{a}(1-\lambda))$, and $(2\pi s^2/|2\pi\upsilon|)^{1/2} = [2\mathfrak{a}(1-\lambda)|\upsilon|]^{-1/2} = [(1-\lambda)\cos\vartheta]^{-1/2}$ by $\mathfrak{a} = \cos\vartheta/(2|\upsilon|)$. The alternative forms of P_λ follow from $\mathfrak{b}/\mathfrak{a} = \tan\vartheta$ and $\operatorname{Re}\upsilon = |\upsilon|\cos\vartheta$. For $\mathfrak{m}_I = 0$ the cross term vanishes and no Young step is needed. □

Only three consequences of Lemma S3.10 are used below: the dominating Gaussian has unit mass (its variance is irrelevant); the penalty factor depends on the leg's starting point only through $\mathfrak{m}_I^2$; and the final bound in Eq. (S95) applies with $\upsilon = \varepsilon^2 v(\zeta)$. For the wrapped transverse kernel the image sum in Eq. (S92) is dominated image by image (there $\mathfrak{m}_I = 0$), and the dominating image sum has unit mass on any period cell.

Lemma S3.11 (Complex-leg kernel data)**.** *Let $h \in (0,1]$ and $z \in \mathbb{C}$ with $u = \operatorname{Re} z > 0$, and set $\zeta = zh$, $A = 2\gamma h u \ge 0$. Then, for the three block variances of Eq.* (S93)*:*

- *(a)* $\operatorname{Re} v(\zeta) > 0$*; specifically* $\operatorname{Re} v_Z(\zeta) \ge D_0 h u\,\mathrm{e}^{-A}$, $\operatorname{Re} v_\parallel(\zeta) \ge 4D_0 h u\,\mathrm{e}^{-A}$, *and* $\operatorname{Re} v_\perp(\zeta) = 4D_0 h u$.
- *(b)* $|\tan\arg v(\zeta)| \le |\operatorname{Im} z|/u$ *for all three blocks.*
- *(c) The mean multipliers obey* $(\operatorname{Im}\mathrm{e}^{-\gamma\zeta})^2 \le \gamma^2 h^2 (\operatorname{Im} z)^2 \mathrm{e}^{-A}$ *for the Z and $\parallel$ blocks; the $\perp$ multiplier is* 1.
- *(d) If additionally $|\operatorname{Im} z| \le r_0$ and $u \ge \tau - r_0$ for some $0 < r_0 < \tau$, then, after Lemma S3.10 with parameter $\lambda \in (0,1)$ is applied blockwise, a Z-block leg of duration ζ started at y_Z carries the penalty factor* $\exp[\omega_Z(h)(y_Z - \bar{z})^2]$ *and a $\parallel$-block leg started at $y_\parallel$ carries* $\exp[\omega_\parallel(h) y_\parallel^2]$, *with*

$$\begin{aligned} \omega_Z(h) &\le \Omega_Z h, \qquad \omega_\parallel(h) \le \Omega_\parallel h, \qquad t_\vartheta = \frac{r_0}{\tau - r_0}, \\ \Omega_Z &= \Big(1 + \frac{t_\vartheta^2}{\lambda}\Big)\frac{\gamma^2 r_0^2}{2\varepsilon^2 D_0(\tau - r_0)}, \qquad \Omega_\parallel = \frac{\Omega_Z}{4}. \end{aligned} \tag{S96}$$

  Both bounds are linear *in the leg fraction h; along any chain whose leg fractions sum to one, the penalty coefficients therefore total at most Ω_Z (respectively $\Omega_\parallel$), independently of the chain order and of the leg configuration. The $\perp$ blocks carry no penalty.*

- *(e) Under the same conditions, the domination constant of one leg (the product over the $d+1$ scalar blocks, the Young step applied only to the Z and $\parallel$ blocks) is at most*

$$\hat{\kappa} = (1-\lambda)^{-1}\big(1 + t_\vartheta^2\big)^{(d+1)/4}. \tag{S97}$$

Proof. (a) $\operatorname{Re}(1 - \mathrm{e}^{-2\gamma\zeta}) = 1 - \mathrm{e}^{-A}\cos(2\gamma h \operatorname{Im} z) \ge 1 - \mathrm{e}^{-A} \ge A\mathrm{e}^{-A}$; multiplying by the positive prefactors in Eq. (S93) gives the two Ornstein–Uhlenbeck bounds, and $\operatorname{Re} v_\perp(\zeta) = 4D_0 h \operatorname{Re} z$ is exact.

(b) For the $\perp$ block, $\arg v_\perp(\zeta) = \arg z$ and $|\tan\arg z| = |\operatorname{Im} z|/u$. For the Ornstein–Uhlenbeck blocks, $|\operatorname{Im}(1 - \mathrm{e}^{-2\gamma\zeta})| = \mathrm{e}^{-A}|\sin(2\gamma h \operatorname{Im} z)| \le \mathrm{e}^{-A}\, 2\gamma h|\operatorname{Im} z| = (|\operatorname{Im} z|/u)A\mathrm{e}^{-A}$; dividing by the real-part lower bound in (a) gives the claim.

(c) $(\operatorname{Im} \mathrm{e}^{-\gamma\zeta})^2 = \mathrm{e}^{-2\gamma h u} \sin^2(\gamma h \operatorname{Im} z) \le \mathrm{e}^{-A} \gamma^2 h^2 (\operatorname{Im} z)^2$.

(d) The Z-block leg kernel is $\mathcal{N}(x_Z - \mathfrak{m}; \varepsilon^2 v_Z(\zeta))$ with $\mathfrak{m} = \bar{z} + (y_Z - \bar{z})\mathrm{e}^{-\gamma\zeta}$, so $\mathfrak{m}_I^2 = (y_Z - \bar{z})^2 (\operatorname{Im} \mathrm{e}^{-\gamma\zeta})^2$. By (b), $|\tan\arg v_Z(\zeta)| \le |\operatorname{Im} z|/u \le t_\vartheta$, so Eq. (S95) and (a), (c) give the penalty exponent

$$P_\lambda \mathfrak{m}_I^2 \le \Big(1 + \frac{t_\vartheta^2}{\lambda}\Big) \frac{\gamma^2 h^2 (\operatorname{Im} z)^2 \mathrm{e}^{-A}}{2\varepsilon^2 D_0 h u \, \mathrm{e}^{-A}} \, (y_Z - \bar{z})^2 = \Big(1 + \frac{t_\vartheta^2}{\lambda}\Big) \frac{\gamma^2 h (\operatorname{Im} z)^2}{2\varepsilon^2 D_0 u} \, (y_Z - \bar{z})^2 \le \Omega_Z h \, (y_Z - \bar{z})^2 :$$

the factors e^{-A} cancel exactly, and $(\operatorname{Im} z)^2 \le r_0^2$, $u \ge \tau - r_0$ were used in the last step. The $\|$ block is identical with $4D_0$ in place of D_0 and mean $y_\|\mathrm{e}^{-\gamma\zeta}$, which yields the extra factor 1/4. The $\perp$ blocks have $\mathfrak{m}_I = 0$ by (c) and the final sentence of Lemma S3.10. Linearity in h and $\sum_k h_k = 1$ give the budget statement.

(e) The Z and $\|$ blocks contribute $[(1-\lambda)\cos\vartheta]^{-1/2}$ each and the $d-1$ transverse blocks $(\cos\vartheta)^{-1/2}$ each, with $|\tan\vartheta| \le t_\vartheta$ throughout by (b), so $\cos\vartheta \ge (1 + t_\vartheta^2)^{-1/2}$ and the product is at most $(1-\lambda)^{-1}(\cos\vartheta)^{-(d+1)/2} \le (1-\lambda)^{-1}(1+t_\vartheta^2)^{(d+1)/4}$. $\square$

After Lemma S3.10 has been applied to every leg of a chain, the surviving dependence on a chain node y is the factor $\exp[\sigma(y_Z - \bar{z})^2 + \sigma' y_\|^2]$ multiplying $V_{w,\varepsilon}(y)$ at interior nodes or $q_0(y)$ at the starting point. The next lemma absorbs these factors by exact Gaussian algebra.

Lemma S3.12 (Penalty absorption by complete squares)**.** *Write* $\phi_w = \sum_j w_j \phi_{j,\varepsilon}$ *for the slab mixture of Eq.* (S34)*,* $\mu_j = \mu(t_j)$ *for its physical centres (distinct from the dimensionless* $c_j = x(t_j)$ *of Eq.* (S24)*), and note* $\sup_y \phi_w = W^{d-1} v_\infty$ *[Eq.* (S36)*; the contact indicator attains* $\sup \chi_a = 1$*].*

(a) (Weighted slab-mixture supremum.) *Let* $\sigma \ge 0$ *with* $x = 2\varepsilon^2\rho^2\sigma < 1$*. For each component the exact complete-square identity*

$$\mathcal{N}(y - \mu_j; \varepsilon^2\rho^2)\, \mathrm{e}^{\sigma(y-\bar{z})^2} = (1-x)^{-1/2} \exp\left[\frac{\sigma(\mu_j - \bar{z})^2}{1-x}\right] \mathcal{N}\Big(y - \mu_j(x); \frac{\varepsilon^2\rho^2}{1-x}\Big), \qquad \mu_j(x) = \frac{\mu_j - x\bar{z}}{1-x}, \tag{S98}$$

holds. Consequently, with $\hat{c} = \max_j |\mu_j - \bar{z}|$*,*

$$\sup_y \big[\phi_w(y)\, \mathrm{e}^{\sigma(y-\bar{z})^2}\big] \le (1-x)^{-1/2} \exp\left[\frac{\sigma \hat{c}^{\,2}}{1-x}\right] S_w(x), \qquad S_w(x) = \sup_y \sum_{j=1}^m w_j \, \mathcal{N}\Big(y - \mu_j(x); \frac{\varepsilon^2\rho^2}{1-x}\Big) : \tag{S99}$$

$S_w(x)$ *is the supremum of the same mixture with every component width inflated by* $(1-x)^{-1/2}$ *and every centre moved outward from* $\bar{z}$ *by the map* $\mu_j \mapsto \mu_j(x)$*. With* $x_V = 2\varepsilon^2\rho^2\Omega_Z$ *(an upper bound for every node argument occurring below), define the inflation slack*

$$\delta_v = \max\Big\{0, \sup_{0 \le x \le x_V} \frac{S_w(x)}{W^{d-1} v_\infty} - 1\Big\} \;\ge\; 0. \tag{S100}$$

(At all three evaluation points of Remark S3.16, the floating-point search gives $\widehat{\delta}_v = 0$*: the width inflation lowers the sampled mixture supremum. This observation is not an interval enclosure of the exact supremum.)*

(b) (Contact node.) *On the support of* χ_a*,* $|y_\|| \le |y|_{\mathrm{mi}} < a$*, so* $\mathrm{e}^{\sigma' y_\|^2} \le \mathrm{e}^{\sigma' a^2}$ *there; the transverse blocks carry no penalty.*

(c) (Initial law.) *For* $\sigma \ge 0$ *with* $x_Z = \varepsilon^2 D_0 \sigma/\gamma < 1$ *and* $\sigma' \ge 0$ *with* $x_\| = 2\varepsilon^2 u_0^2 \sigma' < 1$*,*

$$\int_{\mathbb{R}} \mathcal{N}\Big(y - z_0; \tfrac{\varepsilon^2 D_0}{2\gamma}\Big) \mathrm{e}^{\sigma(y-\bar{z})^2}\, \mathrm{d}y = (1-x_Z)^{-1/2} \exp\left[\frac{\sigma(z_0 - \bar{z})^2}{1-x_Z}\right], \qquad \int_{\mathbb{R}} \mathcal{N}(y - r_{\|,0}; \varepsilon^2 u_0^2)\, \mathrm{e}^{\sigma' y^2}\, \mathrm{d}y = (1-x_\|)^{-1/2} \exp\left[\frac{\sigma' r_{\|,0}^2}{1-x_\|}\right], \tag{S101}$$

exact Gaussian identities; the wrapped transverse initial factor integrates to one against the penalty-free dominating image sums.

(d) (Budget bookkeeping.) *Define*

$$x_V = 2\varepsilon^2\rho^2\Omega_Z, \qquad x_{Z,0} = \frac{\varepsilon^2 D_0\Omega_Z}{\gamma}, \qquad x_{\|,0} = 2\varepsilon^2 u_0^2\Omega_\|, \\ x_{\max} = \max\bigl(x_V,\, x_{Z,0},\, x_{\|,0}\bigr), \tag{S102}$$

and assume $x_{\max} < 1/2$. *(*$x_{\max}$ *is the maximum over* all *complete-square arguments: for parameter sets with* $D_0/(2\gamma) > \rho^2$ *the initial-law argument* $x_{Z,0}$ *exceeds the node argument* x_V*.) Define* $\hat r = \max(a, |r_{\|,0}|)$ *and set*

$$\Pi = \frac{\Omega_Z \hat y^{\,2} + \Omega_\| \hat r^{\,2}}{1 - x_{\max}} + \frac{x_V + x_{Z,0} + x_{\|,0}}{2(1 - x_{\max})}, \qquad \hat y = \max\bigl(|z_0 - \bar z|,\ \max_j |\mu(t_j) - \bar z|\bigr). \tag{S103}$$

Then, along any chain whose leg fractions sum to one, the product of every exponential and normalization factor generated by (a)–(c) is at most e^{Π}*, uniformly in the chain order and the leg configuration.*

Proof. Equation (S98) and both identities in Eq. (S101) follow by completing the square in y in the combined exponent (the second display of (c) is the first with variance $\varepsilon^2 u_0^2$, centre $r_{\|,0}$, and penalty centre 0); they are also verified symbolically in the numerics script. Equation (S99) follows by applying Eq. (S98) componentwise and bounding each centre by $\hat c$. For (b), note that the minimum-image norm dominates the longitudinal component.

(d) Leg k of the chain carries coefficients $\sigma_k = \omega_Z(h_k)$ and $\sigma_k' = \omega_\|(h_k)$, and its penalty lands on the object at its starting point: an interior node — apply (a) and (b) with node argument $x_k = 2\varepsilon^2\rho^2\sigma_k \le x_V$ — or the initial law — apply (c) with arguments at most $x_{Z,0}$, $x_{\|,0}$. Every midpoint centre obeys $(\mu_j - \bar z)^2 \le \hat y^{\,2}$ and $(z_0 - \bar z)^2 \le \hat y^{\,2}$; every longitudinal centre term is at most $\sigma_k' a^2 \le \sigma_k' \hat r^{\,2}$ at interior nodes ($|y_\|| < a$) and, for the leg starting at the initial law, at most $\sigma_k' r_{\|,0}^2/(1 - x_\|) \le \sigma_k' \hat r^{\,2}/(1 - x_{\max})$. The whole-window contact condition controls the deterministic relative trajectory only on I; no condition on $|r_{\|,0}|$ at time zero is used here. By Lemma S3.11(d), $\sum_k \sigma_k \le \Omega_Z$ and $\sum_k \sigma_k' \le \Omega_\|$, and every denominator obeys $1 - x \ge 1 - x_{\max}$, so the exponential factors multiply to at most $\exp[(\Omega_Z \hat y^{\,2} + \Omega_\| \hat r^{\,2})/(1 - x_{\max})]$. For the normalization factors, $\log[1/(1-x)] \le x/(1-x) \le x/(1 - x_{\max})$ gives $\prod_k (1 - x_k)^{-1/2} \le \exp[\sum_k x_k/(2(1 - x_{\max}))] \le \exp[x_V/(2(1 - x_{\max}))]$ (using $\sum_k x_k = 2\varepsilon^2\rho^2 \sum_k \sigma_k \le x_V$), and the two initial-law factors contribute at most $\exp[(x_{Z,0} + x_{\|,0})/(2(1 - x_{\max}))]$. Multiplying proves the claim. ☐

Proposition S3.13 (Flat Dyson chain bound at complex time)**.** *Fix* $t \in I$, $\lambda \in (0,1)$*, and* $0 < r_0 \le \tau/2$ *such that* $x_{\max} < 1/2$ *[Eq.* (S102)*]. Then* $z \mapsto \mathcal{R}_B(z)$ *is analytic on* $\{\operatorname{Re} z > 0\}$*, and on the closed disk* $D_t = \{|z - t| \le r_0\}$*,*

$$|\mathcal{R}_B(z)| \le \hat\kappa v_\infty \mathrm{e}^{\Pi}\bigl(\mathrm{e}^{B v_{\mathrm{eff}}|z|} - 1\bigr), \qquad v_{\mathrm{eff}} = \hat\kappa(1 + \delta_v)v_\infty, \tag{S104}$$

with $\hat\kappa$, Π, δ_v *from Eqs.* (S97), (S103), (S100)*. In particular*

$$\sup_{t\in I}\ \sup_{|z-t|\le r_0} |\mathcal{R}_B(z)| \le E_{\mathrm{c}}(B) = \hat\kappa v_\infty \mathrm{e}^{\Pi}\bigl[\mathrm{e}^{B v_{\mathrm{eff}}(T + r_0)} - 1\bigr]. \tag{S105}$$

Proof. Step 1: analyticity and the Dyson series (the only, qualitative, use of the weighted space). By Proposition S1.1, in the self-adjoint representation $q = \pi u$ the free generator is nonpositive self-adjoint on $L^2(\pi\mathrm{d}x)$ and $-BM_{V_{w,\varepsilon}}$ is a bounded self-adjoint perturbation, so spectral calculus defines both semigroups as analytic operator families on $\{\operatorname{Re} z > 0\}$; conjugating by the unitary multiplication by π gives analytic families on $X_\pi = L^2(\pi^{-1}\mathrm{d}x)$. Since $q_0 \in X_\pi$ [the strict variance inequalities in Eqs. (S19)–(S21)] and $V_{w,\varepsilon} \in L^2(\pi\mathrm{d}x) \cap L^\infty$, the pairing (S91) is continuous and $z \mapsto \mathcal{R}_B(z)$ is analytic on the right half-plane. The norm-convergent Dyson–Phillips series [Eq. (S13), conjugated back from the self-adjoint representation] gives, after pairing,

$$\mathcal{R}_B(z) = -\sum_{n\ge1}(-Bz)^n T_n(z), \qquad T_n(z) = \int_{\Delta_n} \mathcal{T}_n(z;\boldsymbol{s})\,\mathrm{d}\boldsymbol{s}, \\ \mathcal{T}_n(z;\boldsymbol{s}) = \bigl\langle V_{w,\varepsilon},\, \mathrm{e}^{z(1-s_1)\mathcal{L}} V_{w,\varepsilon} \mathrm{e}^{z(s_1-s_2)\mathcal{L}} V_{w,\varepsilon} \cdots V_{w,\varepsilon} \mathrm{e}^{z s_n \mathcal{L}} q_0 \bigr\rangle, \tag{S106}$$

with $\Delta_n = \{1 \ge s_1 \ge \cdots \ge s_n \ge 0\}$, $|\Delta_n| = 1/n!$. No quantitative constant of the X_π theory — neither $\|q_0\|_X$ nor v_*, κ_π — enters any estimate below.

Step 2: flat form, complex continuation, and the identity bridge. Fix n and $\boldsymbol{s}$ in the interior of Δ_n (the boundary is Lebesgue-null and does not affect T_n). Write $h_0 = 1 - s_1$, $h_1 = s_1 - s_2, \ldots, h_n = s_n$ (all positive, $\sum_k h_k = 1$), $\zeta_k = z h_k$, and define the flat chain integral

$$C_n(z;\boldsymbol{s}) = \int\cdots\int V(x_{n+1})\, p_{\zeta_0}(x_{n+1}|x_n)\, V(x_n)\, p_{\zeta_1}(x_n|x_{n-1}) \times \\ \cdots \times V(x_1)\, p_{\zeta_n}(x_1|x_0)\, q_0(x_0)\,\mathrm{d}x_0 \cdots \mathrm{d}x_{n+1}, \tag{S107}$$

with $V = V_{w,\varepsilon}$ and the complex-leg kernels of Eq. (S92). We claim $\mathcal{T}_n(z;\boldsymbol{s}) = C_n(z;\boldsymbol{s})$ on D_t; note $D_t \subset \{\operatorname{Re} z \geq \tau - r_0 \geq \tau/2 > 0\}$. (i) For real $z = t' \in (t - r_0, t + r_0) \subset (0,\infty)$ all kernels are positive, $V \geq 0$, $q_0 \geq 0$; the operator composition in $\mathcal{T}_n$ coincides with the iterated kernel integral by the Markov property of the free semigroup, and Tonelli's theorem identifies the iterated integral with Eq. (S107), the $(n+1)$-point free exposure correlation $\mathbb{E}_0[V(X_{t's_n}) \cdots V(X_{t's_1})V(X_{t'})]$. (ii) For every $z \in D_t$, Lemmas S3.10–S3.12 dominate the integrand of Eq. (S107) in absolute value by a z-independent integrable function (for the fixed leg fractions h_k, every dominating Gaussian and image sum of Lemma S3.10 has variance bounded below on D_t by Lemma S3.11(a), hence is bounded by a constant; the per-leg penalties are at most $\Omega_Z h_k \leq \Omega_Z$ and $\Omega_\| h_k \leq \Omega_\|$ by Lemma S3.11(d); so the integrand is dominated by $C\prod_{k=1}^{n+1} V(x_k) \prod_{k=0}^{n} \mathrm{e}^{\Omega_Z (x_{k,Z}-\bar{z})^2 + \Omega_\| x_{k,\|}^2}\, q_0(x_0)$, which is z-independent and integrable because $x_V, x_{Z,0}, x_{\|,0} < 1$, the contact indicator bounds $|x_{k,\|}| < a$, and the transverse torus is compact); the integral therefore converges absolutely, uniformly on D_t, and $C_n(\cdot;\boldsymbol{s})$ is continuous on D_t by dominated convergence. (iii) $C_n(\cdot;\boldsymbol{s})$ is analytic on the interior of D_t: for every fixed configuration $(x_0,\ldots,x_{n+1})$ the integrand is analytic in z there (each block kernel composes the entire functions $\mathrm{e}^{-\gamma\zeta}$ and $v(\zeta)$ with $\mathcal{N}$, well defined because $\operatorname{Re} v(\zeta_k) > 0$ by Lemma S3.11(a); the transverse image sums converge locally uniformly by the same Gaussian domination); it is jointly measurable; and it is dominated as in (ii). Hence, for every closed triangular contour $\mathcal{C} \subset \operatorname{int} D_t$, Fubini's theorem applies to the double integral $\oint_{\mathcal{C}} C_n(z;\boldsymbol{s})\,\mathrm{d}z$ (contour of finite length, majorant independent of z) and gives

$$\oint_{\mathcal{C}} C_n(z;\boldsymbol{s})\,\mathrm{d}z = \int \cdots \int \Big[\oint_{\mathcal{C}} V(x_{n+1})\, p_{\zeta_0}(x_{n+1}|x_n) \cdots p_{\zeta_n}(x_1|x_0)\, q_0(x_0)\,\mathrm{d}z\Big] \mathrm{d}x_0 \cdots \mathrm{d}x_{n+1} = 0$$

by Cauchy's theorem pointwise in the configuration; Morera's theorem then yields analyticity. (iv) $\mathcal{T}_n(\cdot;\boldsymbol{s})$ is analytic on $\operatorname{int} D_t$ as well: the semigroup families are analytic, the composition is a finite product, and the pairing is continuous (Step 1). By (i), $\mathcal{T}_n$ and C_n agree on the real segment $(t - r_0, t + r_0)$, which has an accumulation point in $\operatorname{int} D_t$; the identity theorem gives $\mathcal{T}_n = C_n$ on $\operatorname{int} D_t$, and both sides are continuous up to the boundary circle (for $\mathcal{T}_n$ because D_t is compactly contained in the open half-plane), so the identity holds on all of D_t.

Step 3: the chain bound. Apply Lemma S3.10 blockwise to every leg kernel in Eq. (S107):

$$|p_{\zeta_k}(x\,|\,y)| \leq \hat{\kappa}\,\bar{g}_k\big(x - \mathfrak{m}_R(y)\big)\, \exp\!\big[\omega_Z(h_k)(y_Z - \bar{z})^2 + \omega_\|(h_k)\, y_\|^2\big],$$

where $\bar{g}_k$ is a product of unit-mass real Gaussians and unit-cell-mass image sums and the constants come from Lemma S3.11(d), (e). Insert these dominations (all $V \geq 0$, $q_0 \geq 0$) and integrate from x_{n+1} inward: the x_{n+1} integral is at most v_∞ (unit mass of $\bar{g}_0$); each x_k integral ($k = n,\ldots,1$) is at most $\sup_x[V(x)\mathrm{e}^{\mathrm{penalty}_k(x)}] \leq (1+\delta_v)v_\infty$ times the exponential and normalization factors of Lemma S3.12(a), (b); the x_0 integral is the initial-law factor of Lemma S3.12(c). By Lemma S3.12(d) the product of all exponential and normalization factors is at most e^{Π}, uniformly in n and $\boldsymbol{s}$. Hence

$$|C_n(z;\boldsymbol{s})| \leq \hat{\kappa}^{\,n+1}(1+\delta_v)^n v_\infty^{n+1}\,\mathrm{e}^{\Pi} \qquad (z \in D_t,\ \boldsymbol{s} \in \Delta_n). \tag{S108}$$

Step 4: summation. By Step 2 and Eq. (S108), $|T_n(z)| \leq |\Delta_n| \sup_{\boldsymbol{s}} |C_n(z;\boldsymbol{s})| \leq (1/n!)\,\hat{\kappa} v_\infty \mathrm{e}^{\Pi}[\hat{\kappa}(1+\delta_v)v_\infty]^n$, so Eq. (S106) gives, for $z \in D_t$,

$$|\mathcal{R}_B(z)| \leq \sum_{n\geq 1} \frac{(B|z|)^n}{n!}\,\hat{\kappa} v_\infty \mathrm{e}^{\Pi}\big[\hat{\kappa}(1+\delta_v)v_\infty\big]^n = \hat{\kappa} v_\infty \mathrm{e}^{\Pi}\big(\mathrm{e}^{B v_{\mathrm{eff}}|z|} - 1\big).$$

Since $|z| \leq t + r_0 \leq T + r_0$ on D_t, taking suprema proves Eq. (S105). □

Proposition S3.14 (Fixed-allocation sufficient budget at a fixed noise scale)**.** *Adopt the setting above. Fix $\varepsilon > 0$ such that the free-exposure clock $G = G_{\varepsilon,w}$ of Eq.* (S37) *has exactly m nondegenerate maxima and $m-1$ nondegenerate minima on $I = [\tau, T]$, ordered alternately, and no stationary endpoint (Theorem S3.8 furnishes this for every $0 < \varepsilon < \varepsilon_0$; a direct verification at one concrete ε serves equally). Let $t_1^* < \cdots < t_{2m-1}^*$ denote the stationary points, and let $U_1,\ldots,U_{2m-1} \subset (\tau, T)$ be any pairwise disjoint closed root tubes with $t_i^* \in \operatorname{int} U_i$, on each of which G'' keeps one fixed sign. Define the margin inventory*

$$\mu_2 = \min_{1\leq i\leq 2m-1} \min_{U_i} |G''| > 0, \qquad \mu_1 = \min_{I \setminus \bigcup_i \operatorname{int} U_i} |G'| > 0, \tag{S109}$$

so that μ_1 in particular bounds both endpoint slopes, since $\tau, T \notin \bigcup_i \operatorname{int} U_i$. (Certified positive lower bounds may be used in place of the two minima: the threshold below is monotone increasing in both.) Fix $\lambda = 1/10$ and the Cauchy radius

$$r_0 = \frac{\varepsilon}{\gamma\hat{y}}\sqrt{\frac{D_0\tau}{2}}, \qquad \hat{y} = \max\!\big(|z_0 - \bar{z}|,\ \max_j |\mu(t_j) - \bar{z}|\big), \tag{S110}$$

and assume the two purely geometric radius conditions

$$\text{(R1)}\quad r_0 \le \frac{\tau}{2}, \qquad\qquad \text{(R2)}\quad x_{\max} < \frac{1}{2}, \tag{S111}$$

with t_ϑ, Ω_Z, $\Omega_\|$, $\hat\kappa$ from Eqs. (S96)–(S97) *and x_V, $x_{Z,0}$, $x_{\|,0}$, $x_{\max}$, Π, δ_v from Lemma S3.12 — all explicit functions of the fixed data. Set*

$$\hat M = \min\!\left(r_0\mu_1,\ \frac{r_0^2\mu_2}{2}\right), \qquad B_{\rm cert}(\varepsilon; w, \{U_i\}) = \frac{\log\!\left(1 + \dfrac{\hat M}{\hat\kappa v_\infty e^{\Pi}}\right)}{\hat\kappa(1+\delta_v)v_\infty\,(T+r_0)}. \tag{S112}$$

Then, for every $0 < B < B_{\rm cert}(\varepsilon; w, \{U_i\})$, the budget-rescaled Doi density $\mathcal{F}_{B,\varepsilon,w} = f_{B,\varepsilon,w}/B$ of Eq. (S85) *has exactly m nondegenerate maxima and $m-1$ nondegenerate minima on I, ordered alternately, and no stationary endpoint; multiplication by B changes no stationary point, so the same holds for the physical density $f_{B,\varepsilon,w}$. In particular $B_{\rm top}(\varepsilon;\{w\}) \ge B_{\rm cert}(\varepsilon; w, \{U_i\})$ for every admissible choice of root tubes. No weighted-space constant enters: v_*, κ_π, and $\|q_0\|_X$ do not appear.*

The closed form Eq. (S110) is chosen for statement cleanliness, not optimality: it makes the leading penalty term manifestly $O(1)$ uniformly in ε, namely $\Omega_Z \hat y^2 = (1 + t_\vartheta^2/\lambda)\,\tau/(4(\tau - r_0))$; any r_0 and λ satisfying (R1)–(R2) are admissible in the proof, and per-case optimization moves the threshold only by an $O(1)$ factor (Remark S3.16).

Proof. Step 1: jet bound at the real control point, weighted-norm-free. Under (R1) every closed disk $D_t = \{|z - t| \le r_0\}$, $t \in I$, lies in the open right half-plane, where $G - \mathcal{F}$ extends to the analytic function $\mathcal{R}_B$ (Proposition S3.13). Cauchy's estimate on D_t together with Eq. (S105) gives

$$\sup_{t\in I}\big|\partial_t^r(\mathcal{F}_{B,\varepsilon,w} - G)\big| \le r!\,r_0^{-r}\,E_{\rm c}(B), \qquad r = 1, 2. \tag{S113}$$

Since $V_{w,\varepsilon} = K_{B,w,\varepsilon}/B$ is independent of B [Eq. (S36)], none of $\hat\kappa$, v_∞, δ_v, Π depends on B, so $E_{\rm c}$ is a continuous, strictly increasing function of B with $E_{\rm c}(0) = 0$.

Step 2: the margins suffice. Suppose

$$r_0^{-1}E_{\rm c}(B) < \mu_1 \qquad \text{and} \qquad 2r_0^{-2}E_{\rm c}(B) < \mu_2, \tag{S114}$$

which is equivalent to $E_{\rm c}(B) < \hat M$. Write $\mathcal{F} = \mathcal{F}_{B,\varepsilon,w}$. The complement $I \setminus \bigcup_i \operatorname{int} U_i$ is a union of $2m$ closed intervals on each of which G' is continuous and nonvanishing, with the alternating sign pattern $+, -, +, \ldots, -$ forced by the signature hypothesis: the first stationary point is a maximum, the types alternate, and no endpoint is stationary. By the $r = 1$ case of Eqs. (S113)–(S114) and $|G'| \ge \mu_1$ there, $\mathcal{F}'$ keeps the fixed sign of G' on each complement interval; in particular $\mathcal{F}'$ has no complement zero and neither endpoint is stationary. The two boundary points of each tube U_i belong to the closed complement, so $\mathcal{F}'$ carries opposite signs there. On U_i, the $r = 2$ case gives $|\mathcal{F}'' - G''| < \mu_2 \le |G''|$, so $\mathcal{F}''$ keeps the fixed sign of G'' and is nonvanishing on U_i; hence $\mathcal{F}'$ is strictly monotone on U_i and, having opposite boundary signs, vanishes exactly once there, at a point where $\mathcal{F}'' \ne 0$. That zero is a nondegenerate maximum where $G'' < 0$ and a nondegenerate minimum where $G'' > 0$. Summing over the $2m - 1$ tubes yields exactly m nondegenerate maxima and $m - 1$ nondegenerate minima, alternating with the tubes, and no other stationary point on I. Multiplication by $B > 0$ changes no stationary point, so $f_{B,\varepsilon,w}$ inherits the signature.

Step 3: inversion. $E_{\rm c}(B) < \hat M$ holds if and only if $e^{Bv_{\rm eff}(T+r_0)} - 1 < \hat M/(\hat\kappa v_\infty e^{\Pi})$, that is, if and only if $B < B_{\rm cert}(\varepsilon; w, \{U_i\})$. By strict monotonicity of $E_{\rm c}$, Eq. (S114) holds for every $0 < B < B_{\rm cert}(\varepsilon; w, \{U_i\})$, and Step 2 applies. □

For the concrete slab design, the only norm entering Proposition S3.14 has a closed Gaussian form. First, for $m = 2$, every $z \in \mathbb{R}$ lies at distance at least $\Delta_{\rm phys}/2$ from one of the two slab centres [Eq. (S34)], whence

$$v_\infty \le \frac{\big(\max_j w_j\big)\big[1 + e^{-\Delta_{\rm phys}^2/(8\varepsilon^2\rho^2)}\big]}{\sqrt{2\pi}\,\varepsilon\rho\,W^{d-1}}, \qquad \Delta_{\rm phys} = |\mu(t_1) - \mu(t_2)|; \tag{S115}$$

for general m, the normalization $\sum_j w_j = 1$ gives the cruder $v_\infty \le (\sqrt{2\pi}\,\varepsilon\rho\,W^{d-1})^{-1}$, and a rigorously enclosed supremum of the mixture may be used in place of either bound (the threshold is monotone decreasing in v_∞, so any upper bound is safe).

Remark S3.15 (The weighted-norm route, retained for comparison)**.** *The first quantitative route — the main proposition of earlier versions of this subsection — applies Theorem S2.1 with control multi-index $\alpha = 0$ and time orders $r \in \{1, 2\}$. For the singleton family the affine chart preceding Eq.* (S10) *may be taken constant ($P = 0$), so the complex-tube suprema of Eq.* (S11) *equal the real-point norms exactly, while $\alpha! = 1$ and $\delta^{-|\alpha|} = 1$; the singleton chart also makes the $\alpha = 0$, δ-free reduction of the mixed-jet theorem exact, not merely proof-mined. Equation* (S12) *then reads*

$$\sup_{t\in I}\left|\partial_t^r(\mathcal{F}_{B,\varepsilon,w} - G)\right| \le r!\left(\frac{2}{\tau}\right)^r E_{\mathrm{wt}}(B), \qquad E_{\mathrm{wt}}(B) = v_*\kappa_\pi\left[\mathrm{e}^{\frac{3}{2}Bv_\infty T} - 1\right]\|q_0\|_X, \tag{S116}$$

with $\kappa_\pi = 1$ on $\mathcal{Q}_d^\infty$, $v_* = \|V_{w,\varepsilon}\|_{L^2(\pi\mathrm{d}x)}$, and $\|q_0\|_X = \|q_0\|_{L^2(\pi^{-1}\mathrm{d}x)}$, all evaluated at the real control point. Steps 2 and 3 of the proof of Proposition S3.14 apply verbatim with (E_{c}, r_0) replaced by $(E_{\mathrm{wt}}, \tau/2)$, giving the weighted-norm threshold

$$B_{\mathrm{top}}(\varepsilon;\{w\}) \;\ge\; B^{\mathrm{wt}}_{\mathrm{cert}}(\varepsilon; w, \{U_i\}) = \frac{2}{3v_\infty T}\log\left(1 + \frac{M}{v_*\kappa_\pi\|q_0\|_X}\right), \qquad M = \min\left(\frac{\tau\mu_1}{2},\ \frac{\tau^2\mu_2}{8}\right). \tag{S117}$$

For the slab design the two weighted norms also have closed Gaussian forms. Matching Eqs. (S16)–(S18) *to the forward operator Eq.* (S2) *identifies $D = \varepsilon^2 D_0$, so the reversible density Eq.* (S5) *factorizes as $\pi = \pi_Z \otimes \pi_\parallel \otimes \pi_\perp$ with $\pi_Z = \mathcal{N}(\,\cdot - \bar z; \varepsilon^2 D_0/(2\gamma))$, $\pi_\parallel = \mathcal{N}(\,\cdot\,; 2\varepsilon^2 D_0/\gamma)$, and $\pi_\perp$ the normalized uniform torus factor of density $W^{-(d-1)}$. The three-Gaussian product formula applied to $\int(\sum_j w_j\phi_{j,\varepsilon})^2\pi_Z\,\mathrm{d}z$ gives, exactly,*

$$v_*^2 = W^{-2(d-1)} I_z I_r, \qquad I_z = \sum_{j,k=1}^m w_j w_k\,\mathcal{N}\big(\mu(t_j) - \mu(t_k); 2\varepsilon^2\rho^2\big)\,\mathcal{N}\Big(\tfrac{\mu(t_j)+\mu(t_k)}{2} - \bar z;\ \tfrac{\varepsilon^2\rho^2}{2} + \tfrac{\varepsilon^2 D_0}{2\gamma}\Big), \tag{S118}$$

where $I_r = \mathbb{P}_{\pi_\parallel\otimes\pi_\perp}(|r|_{\mathrm{mi}} < a) \le 1$ is an explicit one-dimensional integral, since $a < W/2$ excludes wrapped images of the contact ball. Because $\operatorname{Var} Z_0$ *equals the stationary midpoint variance [Eqs.* (S19) *and* (S22)*], the midpoint factor obeys the exact identity $\int\mathcal{N}(z - z_0; v)^2/\mathcal{N}(z - \bar z; v)\,\mathrm{d}z = \mathrm{e}^{(z_0-\bar z)^2/v}$ with $v = \varepsilon^2 D_0/(2\gamma)$. Hence, for isotropic $\Sigma_{\perp,0} = \varsigma_\perp^2 I_{d-1}$,*

$$\begin{aligned}
\|q_0\|_X &= \exp\left[\frac{\gamma(z_0 - \bar z)^2}{\varepsilon^2 D_0}\right]\sqrt{J_\parallel J_\perp},\\
J_\parallel &= \frac{s_\parallel}{\sqrt{v_{\parallel,0}(2s_\parallel - v_{\parallel,0})}}\exp\left[\frac{r_{\parallel,0}^2}{2s_\parallel - v_{\parallel,0}}\right],\\
s_\parallel &= \frac{2\varepsilon^2 D_0}{\gamma}, \qquad v_{\parallel,0} = \varepsilon^2 u_0^2,\\
J_\perp &= \left[\frac{W}{2\sqrt{\pi}\,\varepsilon\varsigma_\perp}\sum_{k\in\mathbb{Z}}\mathrm{e}^{-k^2W^2/(4\varepsilon^2\varsigma_\perp^2)}\right]^{d-1}.
\end{aligned} \tag{S119}$$

the factor W per transverse dimension arising because $\pi_\perp$ has density $W^{-(d-1)}$ [Eq. (S5)*]. The strict hypothesis $u_0^2 < 4D_0/\gamma$ is exactly the condition $2s_\parallel - v_{\parallel,0} > 0$ required by this Gaussian integral.*

Structurally, the two thresholds differ in three places. The denominator of the logarithm drops from $v_*\kappa_\pi\|q_0\|_X$ — carrying $\exp[\gamma(z_0 - \bar z)^2/(\varepsilon^2 D_0)]$, *which is $\mathrm{e}^{1601.856}$ at the matched-law anchor — to the $O(1)$ constant $\hat\kappa v_\infty \mathrm{e}^{\Pi}$. The Cauchy radius shrinks from $\tau/2$ to $r_0 \propto \varepsilon$, a polynomial loss (a factor $\tau/(2r_0) = 20$ at the anchor inside $\hat M$). The exponential-rate denominator changes from $\frac{3}{2}v_\infty T$ to $v_{\mathrm{eff}}(T + r_0)$ (at the anchor slightly* smaller, *i.e. favourable). The weighted norm is intrinsic to the route through Theorem S2.1: its Cauchy–Schwarz pairing must be performed in X_π, where the complex-time free semigroup is a bounded analytic family, and a bounded-quotient realization pays comparably through its similarity bound κ_π. The sharpened route touches q_0 only through mass-one integrals and the initial-law factor of Lemma S3.12(c) (≈ 1.29 at the anchor, part of e^{Π}); neither $\|q_0\|_X$ nor $\|q_0\|_\infty$ ever enters. The price of generality is reversed: the weighted-norm route applies to any analytic positive semigroup with $q_0 \in X_\pi$, while the sharpened route is specific to the product Ornstein–Uhlenbeck/torus structure, the Gaussian slabs, the compactly supported contact indicator, and the Gaussian initial law of the present design.*

Remark S3.16 (Matched-law evaluation of both routes; the bound is conservative)**.** *Take $d = 2$, $m = 2$, and the anchor data $\gamma = 1$, $D_0 = 1$, $z_0 = 4$, $\bar z = 0$, $W = 1$, $a = 0.4$, $\rho = 1$, $\ell_0 = 1$, $I = [0.5, 3.5]$, target times $(t_1, t_2) = (1.0, 2.5)$, $w = (1/2, 1/2)$, and $\varepsilon = 0.1$, so that $S_*^2 = D_0/(2\gamma) + \rho^2 = 3/2$ and $\sigma = \varepsilon S_*/\ell_0 = 0.122474$ [Eq.* (S33)*]. To match* *the off-lattice campaign, take $u_0 = 0.3$, $r_{\|,0} = 0.1$, $r_{\perp,0} = 0$, and $\Sigma_{\perp,0} = 0.3^2$. Then $r_*(t) = (0.1\mathrm{e}^{-t}, 0)$, so the whole-window contact margin Eq.* (S26) *holds, for example, with $\eta = 0.3$, and all remaining hypotheses hold ($z_0 \neq \bar{z}$; $u_0^2 < 4D_0/\gamma$; $a < W/2$; interior targets; $w = (1/2, 1/2)$ is the singleton allocation $w_j = 1/m$). The signature hypothesis of Proposition S3.14 at $\varepsilon = 0.1$ is verified directly, in place of the unquantified ε_0: the stationarity function of Eq.* (S80) *changes sign exactly three times on I, with no stationary endpoint, giving*

$$\begin{aligned} t_1^* &= 0.999079 \ \textit{(max)}, \ G = 1.51358, && G'' = -219.096, \\ t_2^* &= 1.491835 \ \textit{(min)}, \ G = 5.30325 \times 10^{-5}, && G'' = +5.94864 \times 10^{-2}, \\ t_3^* &= 2.491885 \ \textit{(max)}, \ G = 1.30711, && G'' = -9.56943. \end{aligned} \tag{S120}$$

Margin inventory. *As root tubes take $U_1 = [0.942793, 1.04644]$, $U_2 = [1.081873, 2.140298]$, and $U_3 = [2.20198, 2.63618]$. The U_2 endpoints are rounded inward from the half-curvature crossings ($1.0818725\ldots$, $2.1402982\ldots$), so the stated curvature bound holds on the printed closed interval. The floating-point tube search gives $\widehat{\mu}_2 = 2.97432 \times 10^{-2}$ (half the valley curvature), while the complement scan gives $\widehat{\mu}_1 = 1.59288 \times 10^{-11}$, attained at the left endpoint τ, where the exact factorization Eq.* (S37) *yields the ℓ_0-free closed form*

$$|G'(\tau)| = a_\varepsilon(\tau)\, w_1 \mathrm{e}^{-D_L^2/(2S_*^2\varepsilon^2)} \left[\frac{v_\tau D_L}{S_*^2\varepsilon^2} + b_\varepsilon(\tau) + r_2\right], \qquad D_L = \mu(\tau) - \mu(t_1), \tag{S121}$$

with $v_\tau = |\mu'(\tau)| = \gamma(z_0 - \bar{z})\mathrm{e}^{-\gamma\tau}$, $b_\varepsilon = \partial_t \log c_{d,\varepsilon}$, and r_2 the second-slab contribution, suppressed by the odds factor $q_2(x(\tau))/q_1(x(\tau)) = \mathrm{e}^{-116.3}$ [notation of Eq. (S43)*]. Numerically $D_L = 0.954605$, the exponent is $D_L^2/(2S_*^2\varepsilon^2) = 30.3757$ (the window opens $D_L/(\ell_0\sigma) = 7.79$ mixture widths before the first clock), $a_\varepsilon(\tau) = 3.21216$ (contact factor $c_{d,\varepsilon}(\tau) = 0.986127$; the minimum of $c_{d,\varepsilon}$ over I is 0.769771, at the right endpoint), $v_\tau D_L/(S_*^2\varepsilon^2) = 154.399$, and $b_\varepsilon(\tau) = -0.0843601$, giving $|G'(\tau)| = 1.59288 \times 10^{-11}$. This endpoint slope is by far the smallest sampled margin: $|G'(T)| = 0.506775$ and the tube-edge slopes exceed* 1.0. Sharpened-route constants. *With $\lambda = 1/10$: $\hat{y} = 4$ (the initial midpoint dominates; $\max_j |\mu(t_j) - \bar{z}| = 1.4715$), and from Eqs.* (S110), (S96), (S97), (S102), (S103),

$$\begin{gathered} r_0 = 0.0125 \ \textit{[(R1): } 0.0125 \leq 0.25\textit{]}, \qquad t_\vartheta = 0.025641, \qquad \hat{\kappa} = \tfrac{10}{9}\left(1 + t_\vartheta^2\right)^{3/4} = 1.111659, \\ \Omega_Z = 1.6131 \times 10^{-2}, \qquad \Omega_\| = 4.0328 \times 10^{-3}, \qquad x_V = 3.226 \times 10^{-4}, \\ x_{Z,0} = 1.613 \times 10^{-4}, \qquad x_{\|,0} = 7.259 \times 10^{-6}, \qquad x_{\max} = 3.226 \times 10^{-4} \ \textit{[(R2) holds]}, \\ \hat{r} = a = 0.4, \qquad \Pi = 0.2590705, \qquad \mathrm{e}^\Pi = 1.2957251. \end{gathered} \tag{S122}$$

The high-precision floating-point mixture search gives $\widehat{v}_\infty = 1.9947114$ (consistent with the closed-form bound 1.994712 *of Eq.* (S115)*), and the same floating-point search gives $\widehat{\delta}_v = 0$. Substitution gives $\widehat{v}_{\mathrm{eff}} = 2.2174388$ and $\hat{\kappa}\widehat{v}_\infty \mathrm{e}^\Pi = 2.8731911$.* Assembly (sharpened route). *In Eq.* (S112)*, $r_0\widehat{\mu}_1 = 1.99110 \times 10^{-13} \ll r_0^2\widehat{\mu}_2/2 = 2.32369 \times 10^{-6}$, so the $r = 1$ endpoint-slope condition binds; $\widehat{M} = 1.99110 \times 10^{-13}$, $v_{\mathrm{eff}}(T + r_0) = 7.78876$, whence*

$$\widehat{B}_{\mathrm{cert}}(\varepsilon = 0.1; w, \{U_i\}) = 8.8974 \times 10^{-15} \qquad (\log_{10} \widehat{B}_{\mathrm{cert}} = -14.0507). \tag{S123}$$

Weighted-norm route at the same anchor. *From Eqs.* (S115), (S118), (S119)*: $\Delta_{\mathrm{phys}} = 4(\mathrm{e}^{-1} - \mathrm{e}^{-5/2}) = 1.143178$, so $v_\infty \leq 1.994712$; $I_z = 1.283047 \times 10^{-2}$, $I_r = 0.742697$, so $v_* = 9.76174 \times 10^{-2}$; and $J_\| = 4.354047$ and $J_\perp = 9.403160$, so $\log \|q_0\|_X = \gamma z_0^2/(\varepsilon^2 D_0) + \frac{1}{2}\log(J_\| J_\perp) = 1601.8561$ — the initial midpoint sits $(z_0 - \bar{z})/(\varepsilon\sqrt{D_0/(2\gamma)}) = 56.6$ stationary standard deviations from the reversible centre. In Eq.* (S117)*, $\tau\widehat{\mu}_1/2 = 3.98221 \times 10^{-12} \ll \tau^2\widehat{\mu}_2/8 = 9.29475 \times 10^{-4}$ (the $r = 1$ condition binds there too), $2/(3v_\infty T) = 0.0954906$ and $\log[M/(v_*\|q_0\|_X)] = -1625.7786$, whence the nominal floating-point evaluation is*

$$\widehat{B}_{\mathrm{cert}}^{\mathrm{wt}} = 8.19 \times 10^{-708} \qquad (\log_{10} \widehat{B}_{\mathrm{cert}}^{\mathrm{wt}} = -707.0867). \tag{S124}$$

Side by side: the log-argument denominator is $v_*\|q_0\|_X \approx 10^{694.7}$ against $\hat{\kappa} v_\infty \mathrm{e}^\Pi = 2.87$; the Cauchy radius $\tau/2 = 0.25$ against $r_0 = 0.0125$; the exponential-rate denominator $\frac{3}{2}v_\infty T = 10.47$ against $v_{\mathrm{eff}}(T + r_0) = 7.789$. The sharpened expression improves on the weighted-norm one by about 693.0 *orders of magnitude — precisely the removal of the weighted norm, at an $O(10)$ polynomial price.* Secondary evaluation points. *(i) Same geometry, $\varepsilon = 0.05$: the signature is again verified directly (three stationary points, none at the endpoints), and the floating-point inventory gives $\widehat{\mu}_1 = 3.43246 \times 10^{-50}$, attained at τ [consistent with Eq.* (S121)*: exponent $D_L^2/(2S_*^2\varepsilon^2) = 121.503$], and $\widehat{\mu}_2 = 7.29571 \times 10^{-15}$ (half the valley curvature; the valley is now e^{-19}-deep). Constants: $r_0 = 0.00625$, $t_\vartheta = 0.012658$, $\hat{\kappa} = 1.111245$, $\mathrm{e}^\Pi = 1.289539$, $\widehat{v}_\infty = 3.989423$, $\widehat{\delta}_v = 0$; the $r = 1$ condition binds and $\widehat{B}_{\mathrm{cert}} = 2.4142 \times 10^{-54}$ ($\log_{10} = -53.6172$). The weighted-norm route pays the midpoint contribution $\mathrm{e}^{6400} = 10^{2779.6}$ alone, so its corresponding* *sufficient expression lies below* 10^{-2780}*: the sharpened route gains more than* 2700 *orders of magnitude here. (ii)* $m = 3$*, targets* $(0.8, 1.6, 2.8)$*, uniform* $w = (1/3, 1/3, 1/3)$*,* $\varepsilon = 0.1$*: exactly five stationary points, alternating, no stationary endpoint (verified directly);* $\widehat{\mu}_1 = 2.05439 \times 10^{-4}$*, attained at* τ *(the window now opens only* $D_L/(\ell_0\sigma) = 5.13$ *mixture widths before the first clock),* $\widehat{\mu}_2 = 0.490309$ *(half the shallowest curvature, the first valley);* $r_0 = 0.0125$*,* $\hat{\kappa} = 1.111659$*,* $\mathrm{e}^{\Pi} = 1.2957251$*,* $\widehat{v}_\infty = 1.329808$ *(*$\max_j w_j = 1/3$ *lowers the mixture supremum),* $\widehat{\delta}_v = 0$*; the* $r = 1$ *condition binds and* $\widehat{B}_{\mathrm{cert}} = 2.5819 \times 10^{-7}$ *(*$\log_{10} = -6.5881$*) — within seven orders of magnitude of the* $O(1)$ *depletion scale.* Operational comparison and asymptotic rate. *The companion off-lattice campaign returns the protocol-defined classifier crossing* $B_{\mathrm{op}} = 7.62$ *at the matched* $m = 2$ *anchor. The nominal sufficient expression is about* 15 *orders of magnitude smaller. This is a diagnostic comparison, not evidence that* B_{op} *estimates* $B_{\mathrm{top}}(\varepsilon; \{w\})$*. The analytic conservatism is dominated by the left-endpoint slope margin* μ_1 *— the window-opening factor* $\mathrm{e}^{-30.4}$*, which* any *sign-preservation argument on the whole window I pays, since the physical depletion mechanism does not care about the tiny endpoint slope — together with the radius factor* r_0 *and* $O(10)$ *bookkeeping. The semigroup side of the ledger is now* $O(1)$*-tight, so further improvement requires a different proof topology (for instance localizing the signature argument to sub-windows on which* $|G'|$ *is not uniformly tiny, or replacing the uniform complement margin by a degree-theoretic count), not a better transfer estimate. Along the fixed anchor family, provided that, as at both evaluated points, the complement minimum* μ_1 *is attained at the left endpoint* τ *and the* $r = 1$ *condition binds, the endpoint-slope closed form Eq.* (S121) *gives the analytic sufficient expression the leading rate*

$$\log B_{\mathrm{cert}}(\varepsilon; w, \{U_i\}) = -\frac{c_B}{\varepsilon^2} + O\left(\log \frac{1}{\varepsilon}\right), \qquad c_B = \frac{D_L^2}{2S_*^2} = 0.30376. \tag{S125}$$

The evaluations at $\varepsilon = 0.1$ *and* 0.05 *give the predicted decade drop* 39.58 *and the observed nominal drop* 39.57*. For the* $m = 3$ *geometry the corresponding coefficient is* 0.13180*. Under the same proviso, so that the leading rate of Eq.* (S125) *persists for all sufficiently small* ε *(the evaluations verify it only at* $\varepsilon = 0.1$ *and* 0.05*), and since Eq.* (S90) *is one-sided, the only conclusion for the singleton topology threshold is*

$$\liminf_{\varepsilon \downarrow 0} \varepsilon^2 \log B_{\mathrm{top}}(\varepsilon; \{w\}) \geq -c_B.$$

No asymptotic equality is claimed for B_{top}*, and no asymptotic relation is claimed for the protocol-defined* B_{op}*. The Cauchy radius* $r_0 \propto \varepsilon$ *contributes only to the logarithmic correction. Per-case optimization of* r_0 *and* λ *changes the nominal expression only by an* $O(1)$ *factor at the displayed points.*

Numerical-evaluation status. *The transfer constants of Proposition S3.14 [Eqs.* (S96)–(S97), (S102)–(S103)*] are exact closed forms, and the complete-square identities of Lemma S3.12 and both norm factorizations [Eqs.* (S115)–(S119)*] are exact Gaussian algebra, verified symbolically. The displayed margin inventory (contact quadrature, stationary roots, tube edges, complement scan) is computed in high-precision quadrature and bisection (40–60 significant digits; the contact factor by adaptive quadrature with a Chebyshev interpolant, and endpoint derivatives cross-checked against direct differentiation). The displayed matched-law values are reproduced by the companion script, but they have not been independently enclosed; this is a high-precision floating-point calculation, not a directed-rounding interval-arithmetic enclosure. Accordingly the hatted values* $\widehat{\mu}_1$*,* $\widehat{\mu}_2$*,* $\widehat{\delta}_v$*, and* $\widehat{B}_{\mathrm{cert}}$ *are reproducible numerical evaluations of the rigorous sufficient-bound formula, not certified decimal inequalities. A certified decimal value would additionally require interval enclosures for the contact quadrature, every stationary root and tube extremum, the complement derivative minimum, and the slab-mixture supremum. The current reproducibility run also stress-tests Lemma S3.10 on* 10^4 *random samples (worst left/right ratio* 0.99999979*), the two complex-leg inequalities on* 4×10^3 *samples (worst ratios* 0.99999815 *and* 0.99999968*), and the complete-square identities (largest mismatch* 2.12×10^{-37}*). These tests are diagnostics, not substitutes for the analytic proofs or for interval enclosure of the displayed constants.*

S4. NUMERICAL METHODS, CONSISTENCY CHECKS, AND SUPPLEMENTARY FIGURES

This section records the off-lattice methodology behind the numerical campaigns of the main text, the complete production configuration table, the threshold-bisection table, and the supplementary figures.

S4.1. Off-lattice integration and consistency checks

The processes of Eqs. (S16)–(S18) are simulated directly, with no spatial lattice: midpoint, longitudinal, and transverse coordinates evolve by Euler–Maruyama steps of size $\Delta t = 10^{-3}$, the transverse coordinates are wrapped to $[0, W)$ with minimum-image contact distance, and initial conditions are drawn from the laws of Eqs. (S19)–(S21).

Doi killing is realized by per-step thinning with probability $1-\exp(-K_{B,\boldsymbol{w},\varepsilon}\,\Delta t)$, evaluated at the current position (computed via `expm1` for numerical accuracy). Random numbers come from counter-based Philox streams with one `SeedSequence`-spawned substream per walker chunk; the production grid uses seed 20260808 and the five campaign streams use seed 20260813, so every run is deterministic. The common model parameters are $\gamma=1$, $D_0=1$, $\bar{z}=0$, $z_0=4$ (stretched to $z_0=8$ for the $m=5$ demonstration), $W=1$, $a=0.4$, $\rho=1$, $u_0=0.3$, $\varsigma_\perp=0.3$, $r_{\parallel,0}=0.1$, $r_{\perp,0}=0$, window $I=[0.5,3.5]$, $t_{\max}=4$, and target times $(1.0,2.5)$ for $m=2$ and $(0.8,1.6,2.8)$ for $m=3$.

Two automated checks accompany every configuration: the per-step kill probability remains in the unit interval, with largest value 5.6×10^{-3} over the seventeen production configurations and their Δt-halving twin and 3.1×10^{-2} over every campaign run (the $m=2$, $\varepsilon=0.05$, $B=8$ threshold probe), and kills plus survivors equal the walker count exactly. Separately, a campaign-level $B=0$ twin run of 2×10^4 walkers for 500 steps accumulates about 10^7 contact steps and produces zero kills. These are implementation sanity checks, not an accuracy certificate. A Δt-halving twin ($\Delta t=5\times10^{-4}$) of the $(m,\varepsilon,B)=(2,0.1,1)$ production configuration reproduces both peak times at bin resolution and both peak prominences to within 0.3% (Table S1, last row), and the $d=3$ spot check passes a per-bin Δt-halving comparison with maximum $|z|=1.67$ over 26 populated bins.

Modality is decided by a fixed operational classifier: window reaction times are histogrammed with bin width 0.02, smoothed with a Gaussian kernel of bandwidth 0.04, and a local maximum of the smoothed density counts as a mode if and only if its topographic prominence (peak height minus contour-base height) exceeds both five covariance-aware smoothed-Poisson standard deviations and 5% of the maximum smoothed height on I. Conditional on the selected contour-base bin, peak and contour base are linear statistics of the raw bin counts C_j; treating the counts as independent Poisson variables gives the prominence variance $\sigma^2_{\rm prom}=\sum_j\left(A_{p,j}-A_{b,j}\right)^2C_j/(N\delta)^2$, where $A_{p,j}$ and $A_{b,j}$ are the edge-normalized Gaussian kernel weights of the peak bin and of the selected contour-base bin, N the walker count, and δ the bin width. Because each walker reacts at most once, the window counts are multinomial across bins and the Poisson form omits a negative covariance term, so $\sigma_{\rm prom}$ is mildly conservative. A peak-only convention $\sigma^2_{\rm peak}=\sum_j A^2_{p,j}C_j/(N\delta)^2$, which omits the base variance and the peak–base covariance, was the gate applied by the simulation driver at run time and is retained in every stored record for audit; it is not the gate of this article. Every verdict, z-value, bracket, and table reported here is the covariance-aware re-evaluation of the stored raw counts by `reclassify_covariance_aware.py`, which first reproduces the stored smoothed densities, prominences, and $\sigma_{\rm peak}$ values to numerical precision (relative tolerance 10^{-10}) and then applies $\sigma_{\rm prom}$ under the unchanged acceptance rule; the jitter replicas, whose counts are not stored, are regenerated deterministically and re-judged by `w3_jitter_covariance_recheck.py`. Across all 606 stored local maxima of the retained campaign the peak-only convention would inflate z of every counted maximum by a factor 1.00–1.55; the covariance term reduces the variance appreciably only for maxima that the classifier does not count (all far below the five-sigma gate), and no stored maximum crosses the five-sigma threshold upward under the covariance-aware statistic. Classifier verdicts are therefore operational; the main text discusses the one scanned column, $(m,\varepsilon)=(3,0.2)$, that sits at the 5% floor. Because bandwidth, prominence rule, seed, walker count, and step size enter this operational definition, results near a boundary are reported as classifier outcomes rather than exact stationary-point counts.

S4.2. Production configuration table

Table S1 lists all seventeen production configurations and the Δt-halving twin (5×10^6 walkers each, $d=2$, seed 20260808) with their classifier verdicts.

S4.3. Threshold-bisection table

Table S2 lists the operational budget crossings of the equal-weight bisection stream (10^6 walkers per probe, six geometric bisection iterations, pass rule: a classifier-significant maximum later than the last semi-analytic free-clock valley). No stored run in any stream counted more than m significant maxima, and on every stored record of the eight bisection columns, where both rules are defined, the exact-count rule of the phase grid and the last-mode rule of the bisection chains return the same verdict; the two operational rules therefore coincide on the retained data.

S4.4. Classifier sensitivity and repeated-run checks

The retained histogram counts allow the operational classifier to be re-evaluated without resimulating trajectories. On the complete 96-cell phase grid we crossed Gaussian bandwidths $h\in\{0.03,0.04,0.05\}$ with relative-prominence floors $p_{\rm rel}\in\{0.03,0.05,0.07\}$, retaining the five-standard-deviation Poisson condition. Ninety-two cells keep their

TABLE S1. Off-lattice production grid. Columns: prescribed m; noise scale ε; installed budget B; allocation $\boldsymbol{w}$; classified mode count; classifier peak times (histogram bin centres; bin width 0.02); fraction of walkers reacting inside $I = [0.5, 3.5]$; prominence of the weakest counted peak in covariance-aware smoothed-Poisson standard deviations. In the $B = 0.25$ recovery row the classifier additionally finds a third, subthreshold local maximum at $t = 2.49$ with 4.8σ prominence and only 2.7% of the maximum smoothed height, below both the five-sigma condition and the 5% floor.

m	ε	B	$\boldsymbol{w}$	modes	peak times	event mass	min. prom. $[\sigma]$
2	0.1	0.5	$(0.5, 0.5)$	2	0.99, 2.41	0.53	417
2	0.1	0.5	$(0.7, 0.3)$	2	0.99, 2.45	0.44	309
2	0.1	1	$(0.5, 0.5)$	2	0.99, 2.37	0.76	545
2	0.1	1	$(0.7, 0.3)$	2	0.99, 2.41	0.68	398
2	0.1	2	$(0.5, 0.5)$	2	0.99, 2.29	0.92	618
2	0.1	2	$(0.7, 0.3)$	2	0.97, 2.35	0.88	448
2	0.2	0.5	$(0.5, 0.5)$	2	0.99, 2.39	0.35	109
2	0.2	0.5	$(0.7, 0.3)$	2	0.99, 2.43	0.29	72
2	0.2	1	$(0.5, 0.5)$	2	0.99, 2.31	0.57	177
2	0.2	1	$(0.7, 0.3)$	2	0.97, 2.35	0.49	113
2	0.2	2	$(0.5, 0.5)$	2	0.97, 2.21	0.79	284
2	0.2	2	$(0.7, 0.3)$	2	0.97, 2.29	0.72	173
3	0.1	1	uniform	3	0.79, 1.57, 2.63	0.74	255
3	0.1	1	$(0.5, 0.3, 0.2)$	3	0.79, 1.59, 2.67	0.66	167
3	0.2	1	uniform	2	0.79, 1.61	0.54	167
3	0.2	1	$(0.5, 0.3, 0.2)$	2	0.79, 1.61	0.48	183
3	0.2	0.25	uniform	2	0.79, 1.65	0.19	45
2[a]	0.1	1	$(0.5, 0.5)$	2	0.99, 2.37	0.76	546

[a] Δt-halving twin at $\Delta t = 5 \times 10^{-4}$; all other rows use $\Delta t = 10^{-3}$.

TABLE S2. Operational classifier crossings $B_{\mathrm{op}}(\varepsilon; m, \boldsymbol{w}_{\mathrm{eq}}, \mathcal{P})$ for equal slab weights and the fixed protocol $\mathcal{P}$. These values are protocol-dependent finite-sample outcomes, not estimates of the theorem threshold B_{top}. Right-censored entries exceeded the scan limit $B = 8$. In the $(3, 0.2)$ column no stored probe certifies the third mode at any scanned budget $B \in [0.125, 8]$, so no operational crossing exists there under the classifier rule (main text, Sec. IV D). The closest candidates reach 4.2–4.4σ at 5.3–7.5% relative prominence with 10^6 walkers, and 3.4σ at 3.5% in the 3×10^6-walker $B = 0.125$ phase-grid cell.

m	ε	B_{op}	bisection bracket	status
2	0.05	6.99	$[6.95, 7.03]$	bisected
2	0.10	7.62	$[7.58, 7.66]$	bisected
2	0.15	> 8	—	right-censored
2	0.20	> 8	—	right-censored
3	0.05	4.15	$[4.13, 4.18]$	bisected
3	0.10	3.57	$[3.55, 3.59]$	bisected
3	0.15	1.55	$[1.54, 1.56]$	bisected
3	0.20	none	—	no certified crossing (classifier floor)

baseline verdict under all nine settings. The four cells in Table S3, all adjacent to the displayed $m = 3$ boundary, change between two and three classified modes for at least one setting. The principal $(m, \varepsilon, B) = (2, 0.1, 1)$ and $(3, 0.1, 1)$ anchors, the five-mode realization, and the $d = 3$ spot check retain their target classifier counts on the tested 7×6 grid of smoothing bandwidths $h \in \{0.02, 0.03, \ldots, 0.08\}$ and relative prominence floors $p_{\mathrm{rel}} \in \{0, 0.01, 0.03, 0.05, 0.07, 0.10\}$ spanning these endpoints.

The 43 stored threshold probes are deliberately concentrated near the decision regions of the eight bisection chains, so their 28/43 setting-sensitive count is not a phase-diagram-wide instability rate. Table S4 reports what those stored probes do support: the span of geometric bracket midpoints among settings that remain bracketed, and the number of the nine settings censored below or above the stored probe range. No setting has a monotonicity violation on its stored chain. Thus the two-decimal values in Table S2 describe only the declared baseline protocol, not classifier-independent

TABLE S3. Phase-grid cells whose operational mode count is not invariant over the nine classifier settings stated in the text. This table identifies classification sensitivity, not uncertainty in an exact continuum root count.

m	ε	B
3	0.050	4.0
3	0.075	4.0
3	0.100	4.0
3	0.175	0.5

TABLE S4. Post-hoc sensitivity of the stored operational-crossing chains to the nine (h, p_{rel}) settings. "Midpoint span" includes only settings bracketed by the stored probes; the last two columns count left- and right-censored settings, respectively.

m	ε	midpoint span	n_{left}	n_{right}
2	0.05	6.17–7.18	0	3
2	0.10	6.17–7.83	0	3
2	0.15	—	0	9
2	0.20	—	0	9
3	0.05	4.15–5.19	4	0
3	0.10	3.08–3.63	0	3
3	0.15	1.55–1.55	3	3
3	0.20	—	9	0

physical constants.

The sixteen verdict-boundary-adjacent $m = 3$ cells marked with an asterisk in the main-text phase diagram were re-run at 3×10^6 walkers; all sixteen reproduced their 10^6-walker classifier verdict. Three independent deterministic seeds (20260814–20260816) at 10^6 walkers were run for each of six reviewer-sensitive configurations. All 18 verdicts agree within each configuration: the $m = 3$ interior anchor gives 3/3/3 modes; the two sides of its $\varepsilon = 0.1$ threshold give 3/3/3 at $B = 3.50$ and 2/2/2 at $B = 3.64$; the two sides of the $\varepsilon = 0.175$ phase boundary give 3/3/3 at $B = 0.5$ and 2/2/2 at $B = 1$; and the $m = 5$ anchor gives 5/5/5. The largest common-index peak-time span is 0.10 at the boundary-adjacent passing cell and at most 0.02 in the other five configurations.

Four targeted Δt-halving comparisons (seed 20260819) use 5×10^5 walkers per resolution. The step size enters the deterministic `SeedSequence` entropy, so the Δt and $\Delta t/2$ samples use independent streams rather than paired common random numbers. The $m = 3$ interior anchor retains 3 modes, the $m = 5$ anchor retains 5, and the baseline $m = 3$, $(\varepsilon, B) = (0.1, 3.57)$ threshold probe retains 2; counted peak times move by at most 0.02. By contrast, the deliberately boundary-adjacent $(m, \varepsilon, B) = (3, 0.175, 0.5)$ cell changes from 3 modes at $\Delta t = 10^{-3}$ to 2 at $\Delta t = 5 \times 10^{-4}$. Its late candidate peak falls from 7.32% to 4.23% of the global maximum and from 7.5 to 4.2 covariance-aware standard deviations, crossing both classifier conditions. Because the streams differ, this one comparison cannot isolate time-step error from sampling fluctuation. For context, the six other stored runs of this cell at $\Delta t = 10^{-3}$ (the 10^6- and 3×10^6-walker phase-grid runs, the three 10^6-walker seed repeats, and the 5×10^5-walker half of this pair) place the late peak at 6.2–7.3% of the global maximum and 7.5–15.3 covariance-aware standard deviations, so the $\Delta t/2$ value is a single low outlier among seven runs; a paired common-random-number or multi-seed $\Delta t/2$ study would be needed to attribute it. Kill-fraction changes are at most 9.86×10^{-4} across the four pairs. The full histogram sufficient statistics and comparisons are supplied in the versioned reproduction archive. These checks expose joint finite-sample/resolution sensitivity near an operational boundary; they do not turn B_{op} into a theorem threshold or establish time-step convergence.

S4.5. Supplementary figures

Figures S1–S3 collect the three supplementary panels referenced from the main text: the $m = 2$ modality phase diagram, the slab-jitter robustness curves, and the $d = 3$ spot check.

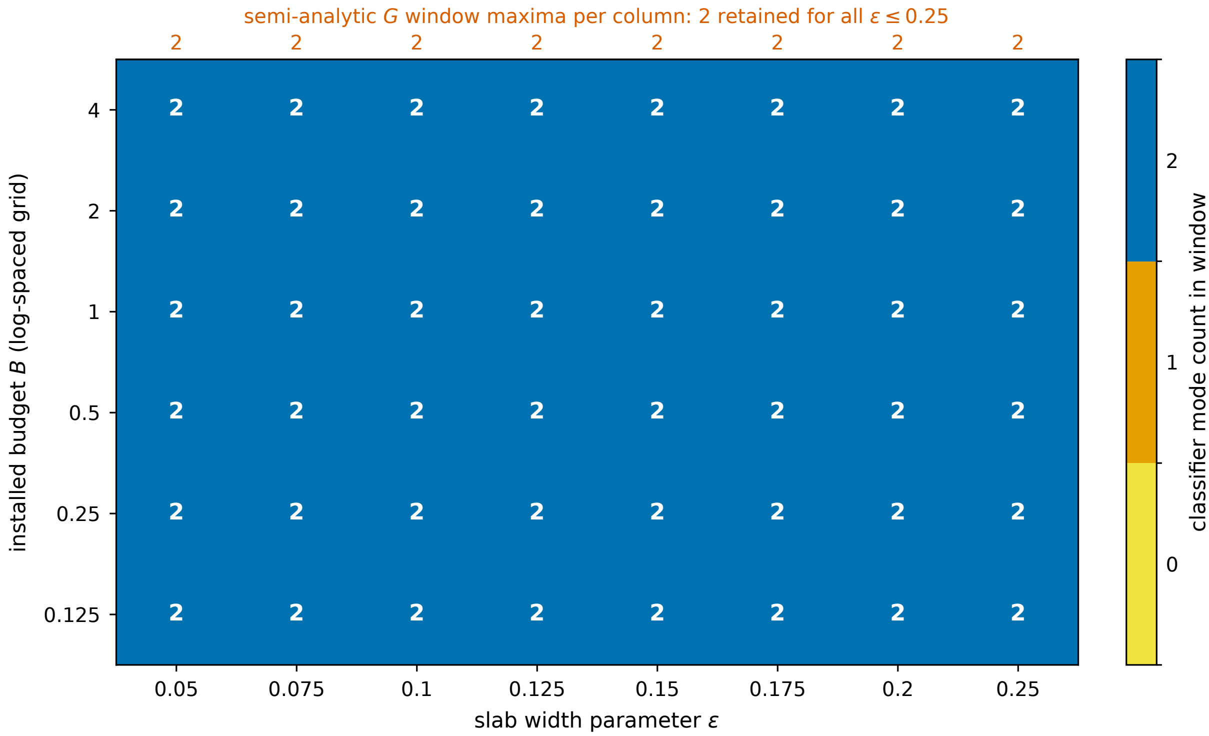

FIG. S1. Observed modality phase diagram for the equal-weight $m = 2$ design: the fixed classifier identifies two significant modes in all 48 (ε, B) cells, and the semi-analytic free clock retains both window maxima throughout the scanned range. (Plotting code revised with Codex assistance; numerical content author-verified.)

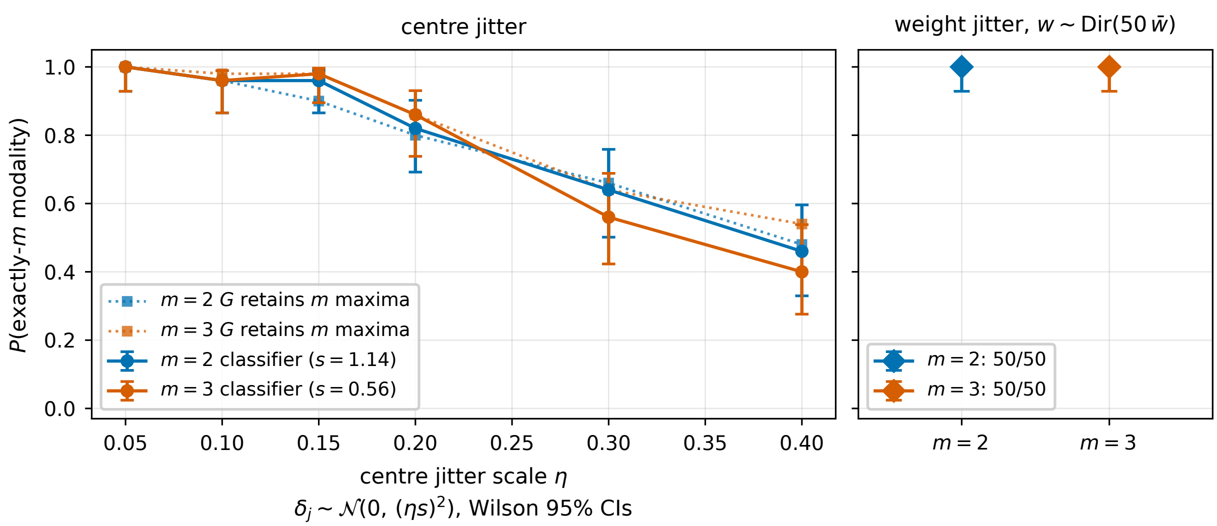

FIG. S2. Design-rule basin under geometry and weight jitter (50 replicas of 2×10^5 walkers per point, Wilson 95% intervals). Left: fraction retaining the target classifier mode count under independent Gaussian centre jitter $\delta_j \sim N(0, (\eta s)^2)$, with s the minimum adjacent slab spacing of the unperturbed design ($s = 1.14$ for $m = 2$, 0.56 for $m = 3$), with the fraction of replicas whose own perturbed semi-analytic clock retains m maxima shown dotted — classifier and perturbed-theory verdicts agree in 86–100% of replicas, a pattern predominantly consistent with geometric peak merger but not excluding sampling-driven mismatches. Right: all 50 weight-jitter replicas $[\boldsymbol{w} \sim \mathrm{Dirichlet}(50\,\bar{\boldsymbol{w}})]$ retain the target classifier mode count for both designs. (Plotting code revised with Codex assistance; numerical content author-verified.)

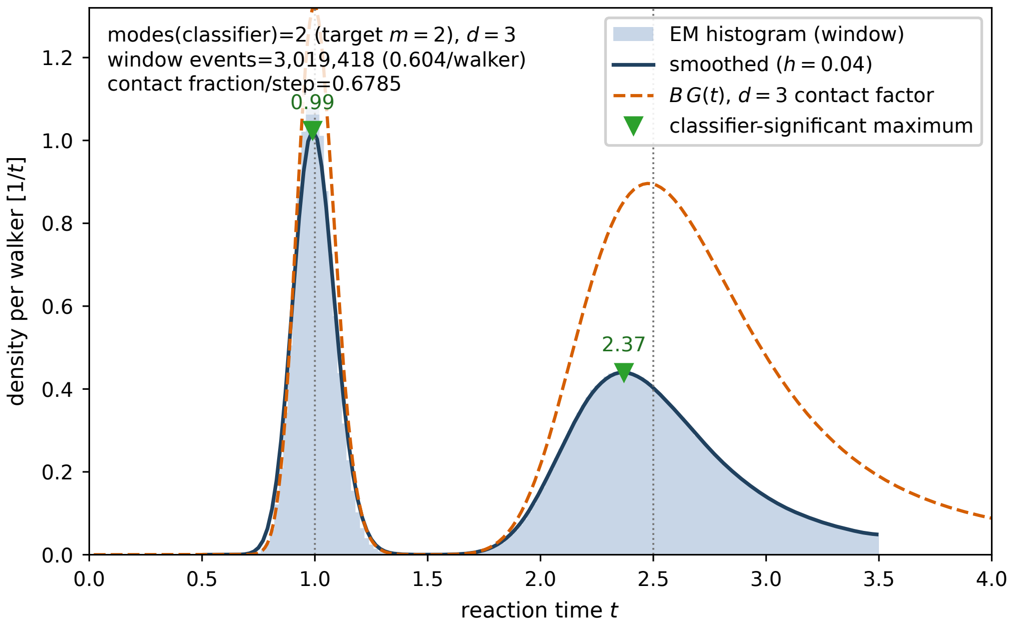

FIG. S3. $d=3$ spot check: two transverse torus directions, killing prefactor B/W^2, all remaining parameters unchanged from the $(m,\varepsilon,B)=(2,0.1,1)$ anchor. The histogram (legend "EM") collects Euler–Maruyama window reaction times. The fixed classifier identifies two significant modes at $t=0.99$ and 2.37, with 0.60 reaction events per walker inside the window and Δt-halving maximum $|z|=1.67$. (Plotting code revised with Codex assistance; numerical content author-verified.)

TABLE S5. Physical-unit conversion of the simulated $d = 2$, $m = 3$ example, with equal weights $w_j = 1/3$, $\varepsilon = 0.1$, $\rho = 1$, $D_0 = 1$, $W = 1$, and $L_0 = 1$ μm. The main column uses $t_0 = 1$ s; parentheses give the alternative $t_0 = 0.01$ s that realizes $D_{\rm phys} = 1$ μm^2/s. Values are indicative floating-point conversions, not a validated parameter set.

Dimensionless quantity	Value used	Physical value	Interpretation
length unit L_0	1	1 μm	reference pattern scale
time unit t_0	1	1 s (0.01 s)	slow-diffusion (unit-diffusivity) anchor
reference coefficient D_0	1	1 μm^2/s (100 μm^2/s)	not the particle diffusivity
particle diffusivity $D = \varepsilon^2 D_0$	0.01	0.01 μm^2/s (1 μm^2/s)	coefficient in the generator
Ornstein–Uhlenbeck rate γ	1	1 s^{-1} (100 s^{-1})	rate scales as t_0^{-1}
contact radius a	0.4	0.4 μm	satisfies $a < W/2$
slab width $\varepsilon\rho$	0.1	100 nm	longitudinal Gaussian width
transverse period W	1	1 μm	identical to the simulated torus
window $[\tau, T]$	$[0.5, 3.5]$	0.5–3.5 s (5–35 ms)	rates and times rescale together
matched-law contact factor	$c(\tau) \simeq 0.986$, $c(T) \simeq 0.770$	dimensionless	floating-point evaluation
budget B	1	$B_{\rm phys} = 1$ μm^2/s (100 μm^2/s)	$B_{\rm phys} = BL_0^d/t_0$
one-slab peak rate at $B = 1$	$[3\sqrt{2\pi}\,\varepsilon\rho W]^{-1} \simeq 1.33$	1.33 s^{-1} (133 s^{-1})	$w_j = 1/3$, $d = 2$
operational classifier crossing	$B_{\rm op} \simeq 3.57$	protocol-dependent	numerical gate only; no theorem claim

S5. CONSISTENT PHYSICAL DIMENSIONALIZATION

Order-of-magnitude laboratory anchor.—The construction is stated in dimensionless form, with particle diffusivity $D = \varepsilon^2 D_0$. Consequently, choosing $L_0 = 1$ μm, $t_0 = 1$ s, $D_0 = 1$, and $\varepsilon = 0.1$ gives the physical diffusivity $D_{\rm phys} = 0.01$ μm^2/s, rather than 1 μm^2/s. This slow-diffusion anchor is appropriate to a confined colloid or a particle in a viscous medium. If instead one fixes $D_{\rm phys} = 1$ μm^2/s while retaining the same dimensionless parameters, then $t_0 = 0.01$ s; all rates below are multiplied by 100 and the window becomes 5–35 ms. Table S5 records both conversions. The dimensionless transverse period is $W = 1$, matching the simulated process; replacing it by $W = 2$ would define a different torus and would require a new numerical evaluation. The decimal entries below are ordinary floating-point conversions, not interval-certified inequalities, and no experimental realization is claimed.

S6. MACHINE-CHECKED KERNELS (LEAN 4)

A Lean 4 package (toolchain `v4.32.0-rc1`, `mathlib4` pinned to the same release) accompanies this article. It machine-checks kernels of the exact-m proof chain—the $2m - 1$ zero bound for the exponential polynomial of Eq. (S47), both in the distinct-zeros form and in the full count with multiplicity of Lemma S3.3, the mixture log-derivative identities of Eq. (S44), and the crossover ratio and nonadjacent smallness bounds with explicit constants—together with selected scalar, Gaussian-integral, and normalization kernels of the explicit budget-threshold proposition (Proposition S3.14), including the two-sided threshold inversion, its continuity, the heterogeneous block-product bound reaching the displayed $(1 - \lambda)^{-1}(1 + t_\vartheta^2)^{(d+1)/4}$ shape, and a uniform $O(1)$ penalty bound under condition (R1). The package was developed through two independent encoding efforts, with duplicate independent encodings for the distinct-zero, mixture-identity, crossover-point, and threshold subkernels. The analytic semigroup, Dyson-expansion, and complex-contour scaffolding of Secs. S1–S3 is *not* formalized; four additional kernels anchor to the fold-transfer theory of a related manuscript by the author [5] and are so labelled in their file headers. The package contains no `sorry`, `admit`, or added axiom. Of its 173 theorem and lemma declarations, the 138 public theorems listed in its `#print axioms` reports report exactly the standard foundation [`propext`, `Classical.choice`, `Quot.sound`]; the remaining declarations are internal proof steps and a smoke test and are not listed separately. Theorem 1 itself is not machine-checked end to end: the budget-threshold kernels take the constants of Proposition S3.14 ($\hat{\kappa}$, v_∞, Π, δ_v, and the margin $\hat{M}$) as abstract real parameters under the positivity hypotheses stated in each theorem, and the block angles of Eq. (S97) as abstract data. The build and the statement-fidelity of every public theorem were additionally audited adversarially in two separate Codex sessions (the computational-tool disclosure of the main text identifies the system); both audit reports and a per-finding resolution record ship with the package, and the residual wording findings of the second round were resolved in the released sources, whose hashes are listed below. Source files and SHA-256 anchors (first 16 hex digits) are listed in Table S6. The package ships with the public archive described in the data-availability statement.

TABLE S6. Lean 4 source files and SHA-256 anchors (first 16 hex digits).

File	SHA-256 (16 hex)
`ExpPolyZeros`	`7d99dc72cfe1cfd8`
`ZeroBound`	`f29cd9d8b637363b`
`MixtureIdentities`	`74a8fc956adfabe2`
`GaussianMixture`	`7e7e60bccd99bc74`
`CrossoverBounds`	`8e1e0816afea792a`
`BudgetThreshold`	`8166ecfc0e2c9390`
`BZeroThreshold`	`8f7dce937bf97771`
`B0ChainKernel`	`5ffaed91d4863c9a`
`WindowSignature`	`5bd7020e5b91a5da`

S7. SCOPE OF THE ANALYTICAL AND NUMERICAL RESULTS

The proof establishes an exact existence theorem for the continuum Doi process. It does not use a finite-volume approximation, and no discretized convergence statement is needed for its conclusion. The off-lattice campaigns of Sec. S4 are time-discrete approximations of the continuum model: no spatial lattice is introduced, but time is discretized, and that time discretization is assessed by the consistency checks of Sec. S4.1, not by a convergence theorem. A separate claim that numerically located stationary points converge, with quantitative rates, to those of the exact process would require discretization-error and sampling-error bounds together with componentwise preservation of the root margins. No such claim is made here.

[1] P. C. Bressloff, Diffusion-mediated absorption by partially-reactive targets: Brownian functionals and generalized propagators, Journal of Physics A: Mathematical and Theoretical **55**, 205001 (2022).

[2] D. S. Grebenkov, Joint distribution of multiple boundary local times and related first-passage time problems with multiple targets, Journal of Statistical Mechanics: Theory and Experiment **2020**, 103205 (2020).

[3] M. A. Carreira-Perpiñán and C. K. I. Williams, On the number of modes of a Gaussian mixture, in *Scale-Space Methods in Computer Vision*, Lecture Notes in Computer Science, Vol. 2695 (Springer, 2003) pp. 625–640.

[4] C. Améndola, A. Engström, and C. Haase, Maximum number of modes of Gaussian mixtures, Information and Inference: A Journal of the IMA **9**, 587 (2020).

[5] X. Zhouyi, Spectral diagnostics and local fixed-budget sensitivity of critical points in finite encounter-reaction models (2026), J. Chem. Phys., submitted (manuscript JCP26-AR-03623).